

MDS220 — Musculoskeletal, CNT & Integumentary System

Hard Clinical Practice Exam · 151 Critical-thinking Questions · Lectures 1–23

Single best answer · Detailed answer key with reasoning at the end

About this paper. Every item is grounded strictly in your uploaded lecture slides and past papers — no outside facts. Questions are written as clinical vignettes and 'differentiation' traps in the style the examiner favours: you must reason from a deficit to the structure, distinguish look-alike lesions, and avoid classification traps. Work the whole paper first, then check the key. Name-only items get short answers.

Section 1 · Connective & Muscular Tissue Histology (Lec 1–3)

1. A pathologist examines granulation tissue from a 4-day-old wound. The dominant fibres stain black with silver and are PAS-positive. As the wound matures over weeks, these fibres are progressively replaced. The mature replacement fibre and its collagen type are:
 - A) Type I collagen replaced by Type III collagen
 - B) Type III collagen replaced by mature Type I collagen
 - C) Elastic fibres replaced by Type II collagen
 - D) Type II replaced by Type IV collagen
 - E) Reticular fibres replaced by Type IV collagen
2. A toluidine-blue section of a lymph node shows a cell with strongly basophilic cytoplasm, an eccentric nucleus with peripheral heterochromatin in a wheel/clock-face arrangement, and a pale juxtanuclear zone. Which statement about this cell is correct?
 - A) It is phagocytic and of monocyte origin
 - B) The pale zone is densely packed with lysosomes
 - C) Metachromatic granules contain stored heparin
 - D) Crystalloid granules cause its acidophilia
 - E) Abundant rough ER produces immunoglobulin
3. During NMJ transmission, a toxin prevents fusion of ACh vesicles with the presynaptic membrane, while a second agent occupies the receptor on the motor end-plate so released ACh cannot act. The pre- and post-synaptic agents respectively are:
 - A) MG antibody presynaptic ; botulinum postsynaptic
 - B) Botulinum presynaptic ; curare postsynaptic
 - C) Tetanus toxin presynaptic ; titin postsynaptic
 - D) Curare presynaptic ; botulinum postsynaptic
 - E) Calmodulin presynaptic ; tropomyosin postsynaptic
4. Which feature most reliably distinguishes a triad of skeletal muscle from a diad of cardiac muscle, as taught for excitation–contraction coupling?
 - A) Triads sit at the Z-line; diads at the M-line
 - B) Triads occur only in smooth muscle fibres
 - C) Diads contain troponin but triads do not
 - D) Triads entirely lack a sarcoplasmic reticulum
 - E) Triad: 1 T-tubule + 2 cisternae at A–I junction
5. A muscle biopsy is graded by myosin-ATPase histochemistry. The fibres staining as Type I dominate. After 6 weeks of limb casting, the patient's muscle wastes. The MAIN cellular mechanism of this immobilisation atrophy is:
 - A) Apoptosis of the motor neurons occurs
 - B) Calcium overload causes widespread necrosis
 - C) Increased protein synthesis enlarges the fibre
 - D) Satellite-cell hyperplasia drives the wasting
 - E) Proteolytic degradation exceeds protein synthesis
6. Heavy resistance training enlarges a muscle. The histological basis taught is:
 - A) An increase in connective tissue alone
 - B) Hypertrophy: more myofibrils per fibre

- C) Hyperplasia: new fibres from satellite cells
- D) Conversion of Type I into smooth muscle
- E) Addition of sarcomeres in series only

7. Which combination correctly matches a muscle type with a UNIQUE regulatory feature, as contrasted in lecture?

- A) Cardiac muscle regenerates via satellite cells
- B) Skeletal muscle: caveolae replace T-tubules
- C) Smooth muscle uses troponin-C for contraction
- D) Smooth muscle: calmodulin/MLCK, dense bodies
- E) Cardiac muscle lacks intercalated discs

8. A polyanionic, non-sulfated, very large glycosaminoglycan forms the backbone of proteoglycan aggregates and gives ground substance its hydrated-gel property. Enzymatic removal of it would MOST directly:

- A) Destroy collagen triple-helix formation
- B) Disrupt assembly of the basal lamina
- C) Reduce water-binding/turgor of ground substance
- D) Abolish the reticular fibre network
- E) Prevent cross-linking of elastin

9. Which collagen-type matching is INCORRECT?

- A) Type IV — main fibre of tendon
- B) Type IV — basal lamina (mesh-forming)
- C) Type I — bone, tendon, dermis
- D) Type III — reticular fibres of spleen/lymphoid organs
- E) Type II — hyaline & elastic cartilage

Section 2 · Cartilage, Bone & Integument (Lec 4–6)

10. A 7-year-old presents with multiple low-impact fractures, blue sclerae, dental fragility and conductive hearing loss. Reasoning from the defective protein, which OTHER tissue finding would you most expect?

- A) Defective elastin in the great vessels
- B) Loss of hyaline cartilage (Type II)
- C) Weak tendons, lax joints and thin skin
- D) Defective basal lamina (Type IV failure)
- E) Loss of reticular fibres (Type III)

11. Articular (hyaline) cartilage of the knee heals poorly after injury. Using the properties taught, which single factor is the MOST important reason?

- A) It is innervated and pain inhibits repair
- B) No perichondrium and avascular (diffusion only)
- C) It is highly vascular and bleeds excessively
- D) Osteoclasts resorb the repair tissue formed
- E) Its matrix is calcified and impermeable

12. In the epiphyseal growth plate, lengthening of the bone occurs because of mitosis in one zone, while chondrocytes die in another. Match correctly:

- A) Length at calcification; death at proliferation
- B) Length at reserve zone; death at proliferation
- C) Length at proliferation; death at calcification
- D) Length at hypertrophy zone; death at reserve
- E) Length at ossification zone; death at reserve

13. A bone-collar forms early in long-bone development. According to the lecture sequence, the bone collar is laid down:

- A) Within the secondary ossification centre
- B) Around the diaphysis during primary ossification
- C) At the epiphyseal line during adulthood

- D) In the marrow cavity after remodelling begins
- E) By intramembranous ossification of the epiphysis

14. An anti-resorptive hormone used in osteoporosis directly inhibits the bone-resorbing multinucleated cell. Identify the cell and its origin correctly:

- A) Chondroclast — of cartilage origin
- B) Osteoblast — of mesenchymal origin
- C) Osteocyte — a trapped former osteoblast
- D) Osteoclast — of haematopoietic origin
- E) Osteoprogenitor — of endosteal origin

15. A hypodermic needle is pushed perpendicularly into the skin of the forearm. Which is the correct order of the FIRST FOUR structures pierced?

- A) Hypodermis → dermis → epidermis → fascia
- B) Epidermis → hypodermis → papillary dermis → reticular dermis
- C) Papillary dermis → epidermis → reticular dermis → hypodermis
- D) Epidermis → papillary dermis → reticular dermis → hypodermis
- E) Epidermis → reticular dermis → papillary dermis → hypodermis

16. Meissner corpuscles (fine-touch) and Pacinian corpuscles (pressure/vibration) lie at different depths. A superficial second-degree burn that abolishes fine touch but spares deep pressure sense has most damaged receptors located in the:

- A) Inside the hair bulb region
- B) Within the stratum corneum layer
- C) Only in the subcutaneous fat layer
- D) The reticular dermis and hypodermis
- E) The dermal papillae (superficial dermis)

17. In the stratum corneum, the protease-inhibitor LEKTI normally restrains KLK proteases that break down corneodesmosomes. With LEKTI loss-of-function, the predicted consequence is:

- A) Secondary vitamin D deficiency
- B) Excess KLK degrades corneodesmosomes early
- C) Failure to shed, with thick retention scale
- D) Markedly increased melanin production
- E) Complete loss of Langerhans cells

18. Which statement about cartilage taught in lecture is TRUE?

- A) Fibrocartilage has a well-defined perichondrium
- B) Elastic cartilage entirely lacks a perichondrium
- C) Discs and menisci are fibrocartilage (hyaline component)
- D) Hyaline cartilage grows by intramembranous ossification
- E) The main collagen of fibrocartilage is Type II

19. Albinism, as described, results from:

- A) Excess eumelanin deposition
- B) Destruction of the basal cell layer
- C) Lack of tyrosinase, so no melanin
- D) Loss of basal keratinocytes
- E) Absence of melanocytes in the skin

Section 3 · Back, Suboccipital Triangle & Vertebral Column (Lec 7–9)

20. A stab wound in the lower back enters through the lumbar (Petit) triangle. From the boundaries taught, which structures form its borders?

- A) Trapezius, scapula, latissimus dorsi
- B) Erector spinae, quadratus lumborum, 12th rib
- C) Serratus post. inferior, iliac crest, oblique
- D) Latissimus dorsi, trapezius, rhomboid major
- E) Latissimus dorsi, external oblique, iliac crest

- 21.** Breath sounds are auscultated over the triangle of auscultation. Its boundaries are:
- A) Latissimus dorsi, trapezius, medial scapula
 - B) Splenius capitis, semispinalis, obliquus
 - C) Trapezius, sternocleidomastoid, the clavicle
 - D) Latissimus dorsi, external oblique, iliac crest
 - E) Rhomboid minor, levator scapulae, scalene
- 22.** During neurosurgery in the suboccipital triangle, the surgeon must protect the vessel running across its floor. Identify the contents and the correct superolateral boundary:
- A) Occipital a.; boundary rectus capitis post. minor
 - B) Greater occipital n. only; boundary trapezius
 - C) Internal carotid a.; boundary semispinalis
 - D) Vertebral a. + C1 n.; superolateral oblique sup.
 - E) Spinal accessory n.; boundary splenius capitis
- 23.** A patient with a deep neck laceration loses sensation but retains motor function of trapezius. Recalling trapezius innervation taught, which is correct?
- A) Motor from accessory (XI); sensory from C3–C4
 - B) Motor from facial nerve; sensory from C2
 - C) Motor and sensory both from accessory nerve
 - D) Motor from C3–C4; sensory from accessory n.
 - E) Motor and sensory from dorsal scapular n.
- 24.** Which structure carries ONLY motor (efferent) fibres, with NO sensory component?
- A) White ramus communicans
 - B) Spinal nerve (mixed)
 - C) Dorsal root
 - D) Ventral (anterior) root
 - E) Dorsal root ganglion
- 25.** A herniated L4–L5 disc compresses a nerve root. Based on the lecture, disc herniations characteristically extend in which direction and most commonly occur at which levels?
- A) Directly anterior; levels C1–C2
 - B) Posterolaterally directed; L4–L5 or L5–S1
 - C) Anteriorly directed; levels T1–T2
 - D) Posterior midline; cervical levels only
 - E) Laterally directed; thoracic levels only
- 26.** A whiplash (severe neck hyperextension) injury most stretches/tears which ligament, and why?
- A) Ligamentum flavum — it resists flexion
 - B) Nuchal ligament — a muscle attachment
 - C) Posterior longitudinal lig. — resists flexion
 - D) Supraspinous lig. — joins spinous tips
 - E) Anterior longitudinal lig. — resists extension
- 27.** Rotation of the head ('NO' movement) occurs mainly at which joint, and which structure holds the pivot in place?
- A) Atlanto-occipital joint; alar ligament
 - B) Median atlanto-axial pivot; transverse ligament
 - C) Zygapophyseal joints; ligamentum flavum
 - D) C2–C3 disc joint; the nuchal ligament
 - E) Atlanto-occipital joint; tectorial membrane
- 28.** Rupture of the transverse ligament of the atlas produces which dangerous result?
- A) Herniation of the C2–C3 disc
 - B) Bony fusion of C1 with C2
 - C) A pure hyperextension injury only
 - D) Freed dens → atlanto-axial subluxation
 - E) Simple loss of cervical lordosis

29. A patient has an exaggerated lateral curvature of the spine. Recalling the lecture definitions, this is:

- A) Lordosis
- B) Scoliosis
- C) Spondylolisthesis
- D) Spina bifida
- E) Kyphosis

30. The intervertebral disc has two parts. The shock-absorbing semifluid core and the surrounding fibrocartilage ring are, respectively:

- A) Synovial pad ; capsule
- B) Anulus fibrosus ; nucleus pulposus
- C) Tectorial membrane ; alar ring
- D) Cartilage end-plate ; ligamentum flavum
- E) Nucleus pulposus ; anulus fibrosus

Section 4 · Scapular, Deltoid, Pectoral & Axilla (Lec 10–11)

31. After axillary lymph-node dissection, a patient cannot adduct/medially rotate the arm well and the lower (sternocostal) part of pectoralis major is weak, but the clavicular head works. Which nerve was injured?

- A) Lateral pectoral nerve
- B) Medial pectoral nerve
- C) Thoracodorsal nerve
- D) Lower subscapular nerve
- E) Long thoracic nerve

32. A patient cannot start abduction of the arm (first ~15°) but, once helped past that, completes abduction normally with the deltoid. Which muscle/nerve is at fault?

- A) Supraspinatus / suprascapular nerve
- B) Deltoid / axillary nerve
- C) Serratus anterior / long thoracic
- D) Trapezius / accessory nerve
- E) Teres minor / axillary nerve

33. Anterior–inferior dislocation of the glenohumeral joint occurs because the capsule is weakest there. Which nerve and artery are most endangered as they traverse the quadrangular space?

- A) Long thoracic n. & lateral thoracic a.
- B) Axillary n. & post. circumflex humeral a.
- C) Musculocutaneous n. & brachial a.
- D) Radial n. & profunda brachii a.
- E) Suprascapular n. & suprascapular a.

34. A winged scapula (medial border lifts off when pushing against a wall) indicates paralysis of which muscle and nerve?

- A) Levator scapulae / C3–C4
- B) Latissimus dorsi / thoracodorsal n.
- C) Rhomboids / dorsal scapular nerve
- D) Serratus anterior / long thoracic n.
- E) Trapezius / accessory nerve

35. Which muscle is supplied by the spinal accessory nerve and, when paralysed, causes shoulder droop and weak scapular elevation?

- A) Levator scapulae
- B) Trapezius
- C) Rhomboid major
- D) Latissimus dorsi
- E) Serratus anterior

Section 5 · Arm, Forearm & Cubital Fossa (Lec 12–13)

36. A phlebotomist draws blood from the median cubital vein. A protective sheet of tissue separates the vein from the underlying brachial artery and median nerve. That structure, and the correct superficial-to-deep order, are:

- A) Interosseous membrane ; vein→nerve→artery
- B) Brachial fascia ; artery→vein→nerve
- C) Flexor retinaculum ; nerve→artery→vein
- D) Bicipital aponeurosis ; vein→it→artery→nerve
- E) Palmar carpal lig. ; artery→nerve→vein

37. A mid-shaft (spiral) humeral fracture causes wrist drop. The nerve and the artery that accompany each other in the spiral groove and may be co-injured are:

- A) Musculocutaneous nerve & ulnar artery
- B) Axillary nerve & post. circumflex a.
- C) Median nerve & brachial artery
- D) Radial nerve & profunda brachii artery
- E) Ulnar nerve & sup. ulnar collateral a.

38. A patient cannot flex the DIP of the index finger or the IP of the thumb and has weak pronation, but sensation is entirely normal. Which nerve is injured and why is sensation spared?

- A) Posterior interosseous nerve
- B) Recurrent branch of median nerve
- C) Ulnar nerve at Guyon's canal
- D) Median nerve at wrist — purely sensory
- E) Anterior interosseous nerve — motor only

39. A laceration posterior to the medial epicondyle. Using forearm innervation taught, which muscles lose function?

- A) Brachioradialis & the supinator muscle
- B) Flexor carpi ulnaris & medial half of FDP
- C) Flexor pollicis longus & pronator quadratus
- D) Flexor carpi radialis & pronator teres
- E) Extensor carpi ulnaris & extensor digitorum

40. A correct forearm muscle–nerve pairing is:

- A) Pronator teres — the ulnar nerve
- B) Extensor carpi ulnaris — median nerve
- C) Flexor pollicis longus — ulnar nerve
- D) Supinator & ECRB — deep radial nerve
- E) Flexor digitorum superficialis — AIN

41. Repetitive wrist flexion in a golfer inflames the common flexor origin. This 'golfer's elbow' involves which structure, and which 'tennis elbow' counterpart is correctly contrasted?

- A) The olecranon ; the coronoid process
- B) Medial epicondyle (flexors) ; lateral (ECRB)
- C) The radial head ; the capitulum
- D) Lateral epicondyle ; medial epicondyle
- E) The trochlea ; the trochlear notch

42. A 'Saturday-night palsy' (radial neuropathy) classically follows compression of the radial nerve against the humerus. The expected deficit is:

- A) Claw hand of ring and little fingers
- B) Loss of medial 1½-finger sensation
- C) Inability to abduct the arm
- D) Wrist drop: lost wrist/finger extension
- E) Loss of thumb opposition

43. A Colles' fracture is described in lecture as:

- A) Distal radius fracture, anterior (volar) displacement
- B) Mid-shaft humeral spiral fracture pattern
- C) Scaphoid waist fracture from a fall

- D) Distal radius fracture, posterior (dorsal) displacement
- E) Olecranon avulsion fracture pattern

44. The brachial artery divides into the radial and ulnar arteries. According to the slides, this bifurcation occurs at:

- A) At the mid-point of the arm
- B) Within the deltopectoral triangle
- C) At the level of the wrist joint
- D) Within the carpal tunnel canal
- E) At the apex of the cubital fossa

Section 6 · Hand & Joints of Upper Limb (Lec 14–15)

45. A woman has wrist trauma with weak thumb flexion, abduction, and inability to oppose the thumb, plus wasting of the thenar eminence — but NO sensory loss anywhere. Which structure is injured?

- A) Median nerve proper at the wrist
- B) Anterior interosseous nerve
- C) Recurrent (motor) branch of median n.
- D) Deep branch of the ulnar nerve
- E) Ulnar nerve at Guyon's canal

46. Carpal tunnel syndrome compresses the median nerve. Recalling the boundaries taught, the flexor retinaculum (transverse carpal ligament) attaches on the ULNAR side to:

- A) The styloid process of the radius
- B) Bases of the 2nd and 3rd metacarpals
- C) The lunate and triquetrum bones
- D) Pisiform bone and hook of hamate
- E) Tubercle of scaphoid and ridge of trapezium

47. A cyclist with prolonged handlebar pressure develops weakness of the interossei, adductor pollicis and hypothenar muscles, with a clawed hand but intact thenar opposition. The lesion is:

- A) Deep ulnar branch in Guyon's canal
- B) Posterior interosseous nerve injury
- C) Radial nerve in the spiral groove
- D) Median nerve within the carpal tunnel
- E) Recurrent branch of the median nerve

48. In an ulnar 'claw hand', the MCP joints are hyperextended and the PIP/DIP joints flexed in the 4th–5th digits. The mechanism, per the lecture, is paralysis of which combination?

- A) Extensor digitorum muscle only
- B) Flexor digitorum superficialis only
- C) The thenar muscle group alone
- D) Lumbricals (medial two) and interossei
- E) Adductor pollicis muscle only

49. Which intrinsic hand muscle is the ODD ONE OUT because it is NOT supplied by the deep branch of the ulnar nerve?

- A) The dorsal interossei
- B) The palmar interossei
- C) Abductor digiti minimi
- D) Opponens pollicis (a thenar muscle)
- E) Adductor pollicis muscle

50. The anatomical snuffbox. From the slides, which set of boundaries/contents is correct?

- A) Medial EPL; lateral EPB/APL; floor scaphoid; radial a.
- B) Medial ED; lateral APL; floor capitate; median n.
- C) Medial EPL; lateral ECRL; floor pisiform; ulnar a.
- D) Medial APL; lateral EPL; floor hamate; ulnar a.
- E) Medial EPB; lateral ECU; floor lunate; ulnar n.

- 51.** Which upper-limb joint-type pairing is INCORRECT, per the summary slide?
- A) Wrist (radiocarpal) — hinge joint
 - B) Proximal radio-ulnar — pivot joint
 - C) Sternoclavicular — saddle (with disc)
 - D) Elbow — a uniaxial hinge joint
 - E) Glenohumeral — ball-and-socket joint
- 52.** A baseball pitcher tears the ulnar collateral ligament of the elbow and undergoes 'Tommy John' reconstruction. The UCL connects which bony points, per the slide?
- A) Olecranon to the head of the radius
 - B) Medial epicondyle to coronoid and olecranon
 - C) Lateral epicondyle to the anular ligament
 - D) Capitulum to the neck of the radius
 - E) Trochlea to the scaphoid bone
- 53.** In a fall on the shoulder tip causing AC joint separation, which ligament — which normally prevents the acromion moving inferiorly when forced medially — is commonly torn along with the AC ligament?
- A) Sternoclavicular ligament tear
 - B) Glenohumeral ligament tear
 - C) Costoclavicular ligament tear
 - D) Coracoclavicular ligament tear
 - E) Coraco-acromial ligament tear
- 54.** 'Nursemaid's elbow' (radial-head subluxation) in a toddler involves which ligament, per the slides?
- A) Radial collateral ligament
 - B) Oblique cord
 - C) Anular ligament of the radius
 - D) Quadratus ligament
 - E) Interosseous membrane

Section 7 · Development of Bone, Skin & Muscle (Lec 16–17)

- 55.** A neonate has a long, narrow, boat-shaped skull. Reasoning from the suture involved, which suture fused prematurely and what is the term?
- A) Sagittal suture — scaphocephaly
 - B) All sutures fused — oxycephaly
 - C) Metopic suture — trigonocephaly
 - D) Coronal (bilateral) — brachycephaly
 - E) Unilateral coronal — plagiocephaly
- 56.** Which skull bone is a CARTILAGINOUS-neurocranium (chondrocranium) derivative, forming by endochondral ossification?
- A) Maxilla
 - B) Parietal
 - C) Sphenoid
 - D) Squamous temporal
 - E) Frontal
- 57.** A child has unilateral absence of the sternocostal head of pectoralis major with ipsilateral hand hypoplasia/syndactyly. This is:
- A) Sprengel deformity
 - B) Klippel–Feil syndrome
 - C) VACTERL
 - D) Poland syndrome
 - E) Prune-belly syndrome
- 58.** Vertebrae and ribs develop from which somite derivative?
- A) Sclerotome
 - B) Myotome

- C) Dermatome
- D) Neural crest
- E) Lateral plate mesoderm

59. Fusion of adjacent digits (failure of separation) is termed:

- A) Cleft hand
- B) Polydactyly
- C) Clubfoot
- D) Syndactyly
- E) Brachydactyly

60. Partial or complete congenital absence of the abdominal wall musculature characterises:

- A) Prune-belly syndrome
- B) Diastasis recti
- C) Gastroschisis
- D) Omphalocele
- E) Poland syndrome

Section 8 · Gluteal Region, Posterior Thigh & Popliteal Fossa (Lec 18–19)

61. A nurse must give a deep IM gluteal injection avoiding the sciatic nerve. Reasoning from the nerve's course, which quadrant is safe and why?

- A) Superolateral — nerve runs lower/medial
- B) Inferolateral — over ischial tuberosity
- C) Superomedial — over the sacrum
- D) Directly over the greater trochanter
- E) Inferomedial — the skin is thinnest

62. A posterior hip dislocation injures the sciatic nerve, producing foot drop. Which division is most vulnerable, and what is lost?

- A) Tibial division — loss of plantarflexion
- B) Obturator nerve — loss of adduction
- C) Sup. gluteal — loss of abduction
- D) Femoral nerve — loss of knee extension
- E) Common fibular div. — dorsiflexion/eversion

63. A bullet enters the popliteal fossa. Recalling the superficial-to-deep order of the central structures, which is the MOST superficial and which the DEEPEST?

- A) Small saphenous vein lies deepest
- B) Vein superficial; tibial nerve deepest
- C) Artery superficial; tibial nerve deepest
- D) Common fibular n. superficial; vein deepest
- E) Tibial nerve superficial; artery deepest

64. Avascular necrosis of the femoral head most often follows damage to which artery, per the lecture?

- A) Inferior gluteal artery
- B) Lateral circumflex femoral a.
- C) Medial circumflex femoral artery
- D) Obturator artery
- E) Profunda femoris artery

65. Which muscle is NOT a 'true hamstring' because it is supplied by the common fibular division of the sciatic nerve rather than the tibial division?

- A) Semitendinosus muscle
- B) All four are true hamstrings
- C) Short head of biceps femoris
- D) Semimembranosus muscle
- E) Long head of biceps femoris

66. Which structure passes through the greater sciatic foramen ABOVE piriformis?

- A) The pudendal nerve
- B) The sciatic nerve
- C) Inferior gluteal nerve
- D) Post. cutaneous n. of thigh
- E) The superior gluteal nerve

67. Gluteus medius and minimus share an action important for gait. Their paralysis (e.g., superior gluteal nerve injury) produces which sign?

- A) Inability to plantarflex the foot
- B) Trendelenburg sign (pelvis drop)
- C) Loss of active knee extension
- D) Loss of active hip adduction
- E) Foot drop on the affected side

68. Gluteus maximus, the chief extensor and lateral rotator of the thigh, is innervated by the:

- A) Pudendal nerve
- B) Inferior gluteal nerve
- C) Sciatic nerve
- D) Femoral nerve
- E) Superior gluteal nerve

Section 9 · Anterior/Medial Thigh, Leg, Foot & Lower-limb Joints (Lec 20–23)

69. To begin flexion of the fully extended, 'locked' (screw-home) knee, one muscle unlocks it by laterally rotating the femur on the fixed tibia. Which one, and what happens if it is torn?

- A) Gracilis — loss of hip adduction
- B) Tibialis posterior — flatfoot results
- C) Soleus — loss of plantarflexion
- D) Popliteus — knee stays locked in extension
- E) Sartorius — loss of knee flexion

70. A fracture at the neck of the fibula injures the most commonly damaged nerve of the lower limb. Identify the nerve and the full deficit:

- A) Sural nerve — a sensory loss only
- B) Superficial fibular only — dorsi intact
- C) Deep fibular only — eversion intact
- D) Tibial n. — loss of plantarflexion
- E) Common fibular n. — foot drop (dorsi+evert)

71. At heel-strike (initial contact) of the gait cycle, the correct joint positions are:

- A) Hip flex, knee extend, plantarflex
- B) Hip flex, knee flex, ankle plantarflex
- C) Hip flex, knee extend, ankle dorsiflex
- D) Hip extend, knee extend, plantarflex
- E) Hip extend, knee flex, ankle plantarflex

72. Obturator-nerve compression in the obturator canal weakens which movement, and which sensory area is affected?

- A) Hip adduction; medial-thigh skin
- B) Ankle dorsiflexion; foot dorsum
- C) Knee extension; anterior-thigh skin
- D) Hip abduction; lateral-thigh skin
- E) Knee flexion; posterior-thigh skin

73. Collapse of the medial longitudinal arch with hindfoot eversion/abduction is which deformity, and dysfunction of which tendon (the dynamic arch support) is a classic adult cause?

- A) Hammer toe — FDL dysfunction
- B) Clubfoot — tibialis anterior cause
- C) Pes planus — tibialis posterior dysfunction

- D) Hallux valgus — adductor hallucis
- E) Claw toes — lumbrical dysfunction

74. The anterior compartment of the leg is supplied by which nerve and artery, and its main action is:

- A) Tibial n., post. tibial a.; plantarflex
- B) Sural n., dorsalis pedis; toe flex
- C) Common fibular, popliteal a.; invert
- D) Deep fibular n., ant. tibial a.; dorsiflex
- E) Superficial fibular, fibular a.; evert

75. The lateral compartment of the leg (fibularis longus and brevis) is supplied by which nerve and produces which movement?

- A) Deep fibular nerve; dorsiflexion
- B) Sural nerve; sensory only
- C) Common fibular trunk; dorsiflex only
- D) Tibial nerve; foot inversion
- E) Superficial fibular n.; eversion

76. Tarsal-tunnel syndrome compresses a nerve behind the medial malleolus, deep to the flexor retinaculum. Identify the nerve and a resulting deficit:

- A) Sural nerve; lateral-foot numbness
- B) Deep fibular n.; first web-space numb
- C) Tibial n.; sole numbness, weak intrinsic
- D) Saphenous n.; medial-leg numbness
- E) Superficial fibular n.; loss of eversion

77. The deep posterior compartment of the leg contains popliteus, FDL, FHL and tibialis posterior, all supplied by the tibial nerve. Tibialis posterior's main action is:

- A) Extension of the knee
- B) Eversion of the foot only
- C) Extension of the toes
- D) Plantarflexion with inversion (arch)
- E) Dorsiflexion together with eversion

78. The hip joint is a multiaxial ball-and-socket joint. Which ligament is the strongest, limits hyperextension, and has a Y-shape attaching to the AIIS?

- A) Transverse acetabular ligament
- B) Ligament of the head of femur
- C) Pubofemoral ligament
- D) Ischiofemoral ligament
- E) Iliofemoral ligament (Y-shaped, AIIS)

79. A 'posterior drawer' sign (tibia displaces posteriorly on the fixed femur) indicates injury to which ligament, and where does it attach on the tibia?

- A) Tibial collateral ligament
- B) PCL — posterior intercondylar area
- C) ACL — anterior intercondylar area
- D) The patellar ligament
- E) Fibular collateral ligament

80. Which knee meniscus is C-shaped, broader posteriorly, and firmly adherent to the deep surface of the tibial collateral ligament — making it more often injured?

- A) The medial meniscus
- B) The lateral meniscus
- C) Neither attaches to a collateral
- D) Both are affected equally
- E) The transverse ligament

81. The acetabulum is formed by the fusion of which three bones at the triradiate cartilage?

- A) Ilium, ischium, pubis

- B) Sacrum, ilium, ischium
- C) Ilium, sacrum, pubis
- D) Sacrum, pubis, ischium
- E) Ischium, pubis, coccyx

82. The great saphenous vein, per the slides, ends by emptying into the:

- A) The dorsal venous arch
- B) The small saphenous vein
- C) The popliteal vein directly
- D) The external iliac vein directly
- E) The femoral vein (saphenous opening)

83. The ankle (talocrural) joint is most stable and 'close-packed' in which position, given the talus geometry taught?

- A) Plantarflexion (narrow talus engaged)
- B) Inversion of the foot
- C) Eversion of the foot
- D) Dorsiflexion (wide talus wedged in)
- E) Neutral with the toes flexed

Section 10 · Integrative Histology, Cartilage & Bone — Challenge Round

84. A researcher stains a trachea section. The cartilage shows chondrocytes in lacunae, a basophilic territorial matrix, and a perichondrium; collagen is not visible by routine H&E because its refractive index matches the matrix. The cartilage type and its 'hidden' collagen are:

- A) Elastic cartilage; Type II collagen
- B) Hyaline cartilage; Type I collagen
- C) Fibrocartilage; Type I collagen
- D) Hyaline cartilage; Type II (masked)
- E) Fibrocartilage; Type II collagen

85. Two students argue about appositional vs interstitial cartilage growth. Which statement is correct?

- A) Interstitial growth adds cells at the surface
- B) Articular cartilage grows only interstitially
- C) Appositional growth needs no perichondrium
- D) Appositional: chondrocytes divide in lacunae
- E) Interstitial: division within the matrix

86. During fracture repair, the stages must occur in order. Which sequence is correct?

- A) Haematoma→bony→soft callus→remodel
- B) Haematoma→soft callus→bony callus→remodel
- C) Soft callus→haematoma→remodel→bony
- D) Bony callus→haematoma→remodel→soft
- E) Remodel→haematoma→bony→soft callus

87. Pathologic loss of bone mass that reduces strength and predisposes to fracture, distinct from defective mineralisation, is:

- A) Osteoarthritis
- B) Osteopetrosis
- C) Osteomyelitis
- D) Osteomalacia
- E) Osteoporosis

88. Which feature does NOT describe an osteoclast?

- A) Resorbs bone at a ruffled border
- B) Responds to calcitonin
- C) Large and multinucleated cell
- D) Of haematopoietic origin
- E) It secretes osteoid matrix

89. Which statement about osteoblasts is FALSE?

- A) They resorb bone via a ruffled border
- B) They form bone by secreting osteoid
- C) They secrete alkaline phosphatase
- D) They have receptors for IGF-1
- E) They become osteocytes when trapped

Section 11 · Upper-limb Nerve Lesions — Differentiation Round

90. Four hand deformities are taught: ape hand, claw hand, hand of benediction, wrist drop. Match each to its lesion. Which pairing is CORRECT?

- A) Benediction (on fist) — median at elbow
- B) Ape hand — ulnar at Guyon's canal
- C) Claw hand — the radial nerve
- D) Wrist drop — median nerve at wrist
- E) Wrist drop — ulnar nerve at the wrist

91. A patient has a claw hand that is MORE pronounced (deeper clawing of ring/little fingers) when the ulnar lesion is at the WRIST than at the elbow. The reason ('ulnar paradox') taught is:

- A) Elbow lesions paralyse lumbricals more
- B) Wrist lesions add a radial-nerve injury
- C) Wrist lesion spares FDP, so DIPs still flex
- D) A wrist lesion also cuts the median nerve
- E) There is no difference in the clawing

92. A surgical-neck-of-humerus fracture causes which deficit, and which test confirms it?

- A) Claw hand; test finger ab/adduction
- B) Lost elbow flexion; test the biceps
- C) Wrist drop; test finger extension
- D) Deltoid palsy + badge numbness; test abduction
- E) Ape hand; test thumb opposition

93. Which cutaneous nerve of the upper limb is NOT derived from the posterior cord?

- A) Post. cutaneous n. of forearm
- B) Post. cutaneous n. of arm
- C) Lat. cutaneous n. of forearm (musculocut.)
- D) Upper lateral cutaneous n. of arm
- E) Lower lat. cutaneous n. of arm

94. For surgery at the SURGICAL NECK of the humerus, which brachial-plexus block is preferred, per the past-paper teaching?

- A) Infraclavicular block
- B) Supraclavicular block
- C) Wrist block
- D) Axillary nerve block
- E) Interscalene block

95. Klumpke palsy (lower-trunk C8–T1) classically produces which hand sign?

- A) Erb 'waiter's tip' posture
- B) Wrist drop
- C) Winged scapula
- D) An ulnar-type claw hand
- E) Deltoid wasting

96. Pain on resisted elbow flexion when the forearm is SUPINATED points most to injury of which muscle?

- A) Pronator teres
- B) Brachioradialis
- C) Brachialis
- D) Coracobrachialis

E) Biceps brachii

Section 12 · Forearm & Hand Detail — Precision Round

97. The superficial flexor muscles of the forearm share a common origin. Which muscle is the ODD ONE because it is supplied by the ULNAR (not median) nerve?

- A) Flexor carpi ulnaris
- B) Pronator teres
- C) Flexor digitorum superficialis
- D) Palmaris longus
- E) Flexor carpi radialis

98. Which deep-layer forearm flexor has a SPLIT nerve supply, with its medial part ulnar and lateral part anterior interosseous (median)?

- A) Flexor digitorum superficialis
- B) Flexor pollicis longus
- C) Pronator teres
- D) Flexor digitorum profundus
- E) Pronator quadratus

99. The 'outcropping' muscles of the thumb (APL, EPB, EPL) emerge from the forearm and are all supplied by which nerve?

- A) Anterior interosseous nerve
- B) Superficial radial nerve
- C) Posterior interosseous nerve
- D) Median nerve
- E) Deep branch of ulnar n.

100. Six osseofibrous tunnels under the extensor retinaculum transmit 12 tendons of nine extensor muscles on the dorsum of the wrist. The retinaculum spans between which two bones?

- A) Scaphoid and lunate
- B) Pisiform and hamate
- C) Radius and 1st metacarpal
- D) Capitate and trapezium
- E) The ulna and radius

101. The lumbricals are unusual muscles. According to the slides, they ORIGINATE from and produce which action?

- A) From carpals; adduct the thumb
- B) From extensor digitorum; abduct fingers
- C) From metacarpals; flex the IP joints
- D) From retinaculum; oppose little finger
- E) From FDP tendons; flex MCP, extend IP

102. Dorsal interossei ABduct and palmar interossei ADduct the fingers relative to the axial line. A useful mnemonic mapping is 'PAD/DAB'. The axial reference line passes through which digit?

- A) Thumb (1st digit)
- B) Index (2nd) finger
- C) Middle (3rd) finger
- D) Ring (4th) finger
- E) Little (5th) finger

103. The deep palmar arch is formed mainly by which artery, and gives rise to which named branches?

- A) Ulnar a.; common palmar digital branches
- B) Radial a. (3rd part); princeps + radialis indicis
- C) Princeps pollicis alone; metacarpals
- D) Median artery; proper digital arteries
- E) Anterior interosseous; recurrent branches

Section 13 · Joints — Classification & Movement Round

104. Match each joint to its type. Which mapping is WRONG?

- A) Proximal radio-ulnar — plane joint
- B) Metacarpophalangeal — condyloid
- C) Sternoclavicular — saddle joint
- D) 1st carpometacarpal — saddle joint
- E) Interphalangeal — hinge joint

105. Movements of the sternoclavicular joint, the only joint linking the upper limb to the axial skeleton, include all EXCEPT:

- A) Elevation and depression
- B) Posterior rotation of the girdle
- C) All listed are SC movements
- D) Pronation/supination of the girdle
- E) Protraction and retraction

106. The glenohumeral joint sacrifices stability for mobility. Which structure deepens the shallow glenoid cavity?

- A) Transverse humeral ligament
- B) Glenoid labrum (fibrocartilage)
- C) The subacromial bursa
- D) Coracohumeral ligament
- E) Coracoacromial ligament

107. In supination, which two muscles are the prime movers, per the radio-ulnar joint slide?

- A) Anconeus and triceps brachii
- B) Pronator teres and pronator quadratus
- C) Supinator and biceps brachii
- D) Flexor carpi radialis and palmaris
- E) Brachioradialis and ECRL

108. The subacromial (subdeltoid) bursa lies between which structures, and its inflammation (bursitis) causes what?

- A) Scapula and chest wall; winging
- B) Humeral head and glenoid; dislocation
- C) Olecranon and skin; elbow swelling
- D) Acromion/deltoid and supraspinatus; arc pain
- E) Clavicle and 1st rib; numbness

Section 14 · Lower-limb Compartments, Vessels & Gait — Applied Round

109. A young man has an anterior-compartment-of-leg crush injury with rising pressure. Reasoning from its contents, the EARLIEST sign of anterior compartment syndrome and the nerve at risk are:

- A) Pain on stretch + web-space numb; deep fibular
- B) Lost knee extension; femoral nerve
- C) Loss of eversion; superficial fibular
- D) Loss of plantarflexion; tibial nerve
- E) Sole numbness; the tibial nerve

110. The popliteal artery is the deepest central structure of the popliteal fossa. It continues as which arteries, per the lecture?

- A) Genicular arteries only
- B) Anterior tibial and posterior tibial
- C) Femoral and profunda femoris
- D) Fibular and lateral plantar
- E) Dorsalis pedis and medial plantar

111. During the stance-to-push-off transition of gait, the triceps surae (gastrocnemius + soleus) act to:

- A) Plantarflex for push-off
- B) Dorsiflex the ankle joint
- C) Invert the foot
- D) Evert the foot
- E) Extend the knee

112. The small saphenous vein and the great saphenous vein terminate, respectively, in the:

- A) Femoral vein ; popliteal vein
- B) Popliteal vein ; femoral vein
- C) Popliteal ; popliteal
- D) Femoral ; external iliac
- E) External iliac ; internal iliac

113. Foot drop from common fibular nerve injury produces a characteristic gait. Which description fits, per the lecture?

- A) A Trendelenburg waddle
- B) Steppage gait with foot-slap
- C) A festinating gait
- D) A scissoring gait
- E) An antalgic painful limp

114. The medial circumflex femoral artery is critical for the femoral head. Which other arteries contribute to the hip joint's blood supply, per the joints lecture?

- A) Popliteal and genicular arteries
- B) Superior gluteal artery alone
- C) Only the obturator artery
- D) Retinacular, lig. teres, circumflex aa.
- E) Dorsalis pedis branches

Section 15 · Spine, Development & Integrative — Final Challenge

115. A weightlifter develops lumbar canal stenosis with neurogenic claudication. Hypertrophy/buckling of which posterior-wall ligament most narrows the canal?

- A) Anterior longitudinal ligament
- B) Nuchal ligament
- C) Posterior longitudinal lig.
- D) Ligamentum flavum
- E) Supraspinous ligament

116. The primary curvatures of the spine (present in the fetus) versus the secondary curvatures (developing after birth) are correctly listed as:

- A) All four are secondary curvatures
- B) Primary cervical only; secondary the rest
- C) Primary thoracic/sacral; secondary cervical/lumbar
- D) Primary cervical/lumbar; secondary thoracic
- E) All four are primary curvatures

117. The dens (odontoid) of C2 is held against the anterior arch of C1 by a ligament stronger than the dens itself. That ligament and the joint type are:

- A) Alar ligament; a gliding joint
- B) Ant. longitudinal lig.; a plane joint
- C) Tectorial membrane; a hinge joint
- D) Nuchal ligament; a saddle joint
- E) Transverse ligament; median pivot joint

118. Which craniovertebral joint provides the 'YES' nod (flexion/extension), and which provides the 'NO' rotation?

- A) Both give YES at C2–C3
- B) Atlanto-axial YES; occipital NO

- C) Atlanto-occipital YES; atlanto-axial NO
- D) Costovertebral YES; sacroiliac NO
- E) Zygapophyseal YES; disc NO

119. Reasoning across systems: a patient with osteoporosis develops an exaggerated thoracic kyphosis ('dowager's hump'). The structural cause taught is:

- A) Anterior wedge fractures of thoracic bodies
- B) Lateral wedging of lumbar bodies
- C) Fusion of the sacral vertebrae
- D) Disc herniation at C1–C2 level
- E) Ossification of the nuchal ligament

120. Spondylolisthesis, contrasted with spondylolysis and spondylitis, is best described as:

- A) Lateral curvature of the spine
- B) A defect/fracture of the pars interarticularis without slippage
- C) Fusion of vertebral bodies
- D) Forward slippage of one vertebral body on the one below
- E) Inflammation of a vertebra

121. A child has bilateral clubfoot (talipes equinovarus). The deformity, per the development lecture, fundamentally involves abnormal position/development of which bone?

- A) The metatarsal bones alone
- B) The cuboid bone alone
- C) The navicular bone alone
- D) The talus (foot inverted/plantarflexed)
- E) The calcaneus bone alone

122. Hypertrichosis is described as resulting from abnormal development of the:

- A) Mammary gland
- B) Sebaceous gland
- C) Sweat gland
- D) Arrector pili muscle
- E) Hair follicle

123. Premature bilateral fusion of the CORONAL sutures produces a short, wide, tall skull termed:

- A) Trigonoccephaly
- B) Oxycephaly
- C) Plagiocephaly
- D) Brachycephaly
- E) Scaphocephaly

124. A femoral hernia, contrasted with an inguinal hernia in the past papers, protrudes:

- A) Below inguinal lig., via femoral canal
- B) Medial to pubic tubercle, above ligament
- C) Lateral to tubercle, above ligament
- D) Through the lumbar triangle
- E) Through the umbilical ring

Section 16 · Mixed Clinical Vignettes — Capstone Round

125. A 22-year-old falls on an outstretched hand and has tenderness in the anatomical snuffbox. Reasoning from the snuffbox floor and its blood supply, which bone is likely fractured and what complication is feared?

- A) Hamate; ulnar nerve injury
- B) Lunate; carpal-tunnel syndrome
- C) Scaphoid; avascular necrosis risk
- D) Pisiform; FCU tendon rupture
- E) Trapezium; thumb arthritis

126. A diabetic has burning sole pain and tingling worse at night, relieved by ankle movement, with a positive tap sign behind the medial malleolus. The compressed nerve and tunnel are:

- A) Sural nerve; the lateral ankle
- B) Saphenous nerve; adductor canal
- C) Deep fibular nerve; anterior ankle
- D) Common fibular nerve; fibular neck
- E) Tibial nerve; the tarsal tunnel

127. A footballer is struck on the LATERAL knee (valgus force). Per the 'unhappy triad' logic and the meniscal attachments taught, which combination is most likely injured together?

- A) PCL, lateral meniscus, fibular collateral
- B) Fibular collateral and ACL only
- C) Patellar ligament and quadriceps tendon
- D) The lateral meniscus alone
- E) Tibial collateral, medial meniscus, ACL

128. A patient cannot extend the knee and has lost the patellar reflex, with anterior-thigh sensory loss. The nerve and compartment are:

- A) Sciatic nerve; posterior compartment
- B) Tibial nerve; posterior leg
- C) Sup. gluteal nerve; gluteal region
- D) Femoral nerve; anterior compartment
- E) Obturator nerve; medial compartment

129. After prolonged lithotomy positioning, a patient has weak hip adduction and numb medial thigh, with intact knee extension. The nerve is:

- A) Femoral nerve
- B) Common fibular nerve
- C) Sciatic nerve
- D) Tibial nerve
- E) Obturator nerve

130. A swimmer with shoulder pain on overhead motion has a painful arc around 60–120° of abduction and weakness initiating abduction. The most likely structure and bursa involved are:

- A) Subscapularis tendon; subscapular bursa
- B) Supraspinatus tendon; subacromial bursa
- C) Long head of biceps; bicipital groove
- D) Teres minor; the quadrangular space
- E) The deltoid muscle; no bursa

131. A cyclist leaning on the handlebars develops wasting of the first dorsal interosseous and cannot cross the fingers, but thumb opposition and sensation over the thenar area are normal. Localise precisely:

- A) Ulnar nerve at the elbow
- B) Median nerve at the wrist
- C) The radial nerve
- D) Anterior interosseous nerve
- E) Deep ulnar branch in Guyon's canal

132. A toddler is lifted by one arm and then refuses to use it, holding the forearm pronated and slightly flexed. The diagnosis and the structure involved are:

- A) Supracondylar fracture; brachial artery
- B) Monteggia fracture; ulnar nerve
- C) Clavicle fracture; subclavian vessels
- D) Shoulder dislocation; axillary nerve
- E) Radial-head subluxation; annular ligament

133. A patient with long-standing carpal tunnel syndrome has thenar wasting and cannot oppose the thumb ('ape hand'). Yet sensation over the central palm is relatively preserved. The anatomical reason for spared central-palm sensation is:

- A) Sensation runs in the deep branch
- B) The ulnar nerve supplies the whole palm
- C) Median has no sensory fibres at wrist
- D) The radial nerve supplies the palm
- E) Palmar cutaneous branch bypasses the tunnel

134. Reasoning from joint type: a 'pivot' synovial joint permits which single movement, and which lower-/upper-limb examples fit?

- A) Flexion/extension; elbow and knee
- B) Multiaxial; the hip and shoulder
- C) Gliding only; intercarpal joints
- D) Rotation; radio-ulnar and atlanto-axial
- E) Abduction/adduction; the MCP joints

135. A marathon runner develops medial-arch pain and visible flattening of the arch when standing. Which structure, the dynamic supporter taught, has likely failed, and which passive structure also supports the arch?

- A) Tibialis posterior; plantar aponeurosis/spring
- B) Gastrocnemius; iliotibial tract
- C) Flexor hallucis longus; patellar lig.
- D) Tibialis anterior; deltoid ligament
- E) Fibularis longus; the ACL

136. A boxer fractures the neck of the 5th metacarpal ('boxer's fracture'). Which intrinsic muscle group, attaching to that region and supplied by the deep ulnar nerve, may be affected in grip?

- A) The pronator quadratus
- B) The thenar muscle group
- C) The lateral two lumbricals
- D) Outcropping thumb muscles
- E) The hypothenar muscle group

137. Integrative: a patient with a high radial-nerve lesion (axilla) has wrist drop AND loss of elbow extension, whereas a mid-shaft (spiral groove) lesion spares elbow extension. The reason is:

- A) Elbow extension is a median function
- B) Spiral-groove lesion cuts the ulnar nerve
- C) Triceps is a musculocutaneous muscle
- D) Axillary lesion also cuts median nerve
- E) Triceps branches arise above the groove

138. A patient cannot abduct the arm beyond $\sim 15^\circ$ and the deltoid is wasted with lateral-arm numbness, after a blow to the shoulder. Trace the lesion: which nerve, from which cord, through which space?

- A) Radial; post. cord; triangular interval
- B) Suprascapular n.; upper trunk; notch
- C) Axillary n.; post. cord; quadrangular space
- D) Long thoracic; roots; the chest wall
- E) Musculocutaneous; lateral cord; coracobr.

139. Final integrative trap: which statement correctly distinguishes the action loss in a TIBIAL nerve injury from a COMMON FIBULAR nerve injury at the knee?

- A) Both cause identical deficits
- B) Tibial $\rightarrow$ foot drop; fibular $\rightarrow$ plantarflex loss
- C) Tibial $\rightarrow$ eversion; fibular $\rightarrow$ inversion loss
- D) Tibial $\rightarrow$ knee extension; fibular $\rightarrow$ adduction
- E) Tibial $\rightarrow$ plantarflex/invert loss; fibular $\rightarrow$ foot drop

Section 17 · Rapid High-Yield Round

140. Carpal-tunnel syndrome compresses which nerve, and which thumb function is classically lost?

- A) Radial nerve; extension

- B) Ulnar nerve; adduction
- C) Median nerve; opposition of the thumb
- D) Musculocutaneous; flexion
- E) Posterior interosseous; abduction

141. Which rotator-cuff muscle is NOT one of the four (SITS) listed in the glenohumeral slide?

- A) Supraspinatus
- B) Subscapularis
- C) Infraspinatus
- D) Teres major
- E) Teres minor

142. The chief flexor of the elbow joint and the chief extensor, per the elbow-joint slide, are respectively:

- A) Brachioradialis; supinator
- B) Biceps; anconeus
- C) Coracobrachialis; triceps
- D) Brachialis; triceps brachii
- E) Pronator teres; biceps

143. Which compartment of the leg, if its nerve is injured, abolishes EVERSION while sparing dorsiflexion?

- A) Gluteal compartment (sciatic)
- B) Anterior compartment (deep fibular)
- C) Lateral compartment (superficial fibular)
- D) Posterior deep compartment (tibial)
- E) Posterior superficial (tibial nerve)

144. The erector spinae, chief extensor of the vertebral column, splits into three columns. From lateral to medial they are:

- A) Spinalis, longissimus, iliocostalis
- B) Iliocostalis, longissimus, spinalis
- C) Splenius, longissimus, spinalis
- D) Longissimus, spinalis, iliocostalis
- E) Multifidus, rotatores, semispinalis

145. The transversospinalis group, deep to erector spinae, from superficial to deep is:

- A) Splenius, semispinalis, multifidus
- B) Multifidus, semispinalis, rotatores
- C) Rotatores, multifidus, semispinalis
- D) Spinalis, longissimus, iliocostalis
- E) Semispinalis, multifidus, rotatores

146. Which gluteal-region muscle, paralysis of which gives a positive Trendelenburg sign, is supplied by the superior gluteal nerve?

- A) Piriformis
- B) Gluteus medius (and minimus)
- C) Obturator internus
- D) Gluteus maximus
- E) Quadratus femoris

147. The thenar muscles (abductor pollicis brevis, opponens pollicis, superficial flexor pollicis brevis) are supplied by which nerve?

- A) Anterior interosseous nerve
- B) Posterior interosseous nerve
- C) Deep branch of ulnar nerve
- D) Superficial radial nerve
- E) Recurrent branch of the median nerve

148. The interosseous membrane between radius and ulna is which kind of fibrous joint?

- A) Suture
- B) Gomphosis

- C) Synchondrosis
- D) Symphysis
- E) Syndesmosis

149. Which intervertebral ligament limits flexion and is elastic, helping straighten the column after bending?

- A) Anterior longitudinal ligament
- B) Tectorial membrane
- C) Ligamentum flavum
- D) Supraspinous ligament
- E) Nuchal ligament

150. Per the gait lecture, inability to 'unlock' the extended knee implicates which muscle?

- A) Sartorius
- B) Soleus
- C) Popliteus
- D) Quadratus femoris
- E) Tibialis posterior

151. The 1st carpometacarpal (thumb) joint, which allows opposition, is which synovial type?

- A) Plane
- B) Ball-and-socket
- C) Saddle
- D) Hinge
- E) Pivot

Answer Key & Detailed Explanations

Reasoning provided for analytical items; short answers for naming items.

Section 1 · Connective & Muscular Tissue Histology (Lec 1–3)

1. A pathologist examines granulation tissue from a 4-day-old wound. The dominant fibres stain black with silver and are PAS-positive. As the wound matures over weeks, these fibres are progressively replaced. The mature replacement fibre and its collagen type are:

Answer: B) Type III collagen replaced by mature Type I collagen

Early wound/granulation tissue is rich in Type III collagen (reticular fibres) — argyrophilic (silver+) and PAS-positive. During maturation/remodelling it is replaced by stronger Type I collagen, which gives the scar tensile strength.

2. A toluidine-blue section of a lymph node shows a cell with strongly basophilic cytoplasm, an eccentric nucleus with peripheral heterochromatin in a wheel/clock-face arrangement, and a pale juxtanuclear zone. Which statement about this cell is correct?

Answer: B) The pale zone is densely packed with lysosomes

The clock-face nucleus identifies a plasma cell. Its cytoplasmic basophilia is due to extensive rough ER (ribosomal RNA) for antibody synthesis; the pale Hof is the Golgi. Heparin granules belong to mast cells; phagocytosis/monocyte origin to macrophages.

3. During NMJ transmission, a toxin prevents fusion of ACh vesicles with the presynaptic membrane, while a second agent occupies the receptor on the motor end-plate so released ACh cannot act. The pre- and post-synaptic agents respectively are:

Answer: B) Botulinum presynaptic ; curare postsynaptic

Botulinum toxin blocks presynaptic ACh release; curare competitively blocks the postsynaptic nicotinic receptor. (Myasthenia gravis is the autoimmune postsynaptic analogue.) Knowing which side each acts on is the discriminator.

4. Which feature most reliably distinguishes a triad of skeletal muscle from a diad of cardiac muscle, as taught for excitation–contraction coupling?

Answer: E) Triad: 1 T-tubule + 2 cisternae at A–I junction

Skeletal muscle has triads (one T-tubule flanked by two SR terminal cisternae) at the A–I junction. Cardiac muscle has diads (one T-tubule + one SR cisterna), typically at the Z-line. Smooth muscle uses caveolae instead.

5. A muscle biopsy is graded by myosin-ATPase histochemistry. The fibres staining as Type I dominate. After 6 weeks of limb casting, the patient's muscle wastes. The MAIN cellular mechanism of this immobilisation atrophy is:

Answer: C) Increased protein synthesis enlarges the fibre

Disuse/immobilisation atrophy results chiefly from proteolytic degradation outpacing synthesis (proteasome-mediated), shrinking fibre diameter — not loss of fibre number. Type I postural fibres are especially affected by disuse.

6. Heavy resistance training enlarges a muscle. The histological basis taught is:

Answer: B) Hypertrophy: more myofibrils per fibre

Skeletal-muscle enlargement from training is hypertrophy — more myofibrils within existing fibres — not hyperplasia (fibre number is essentially fixed in humans). This is the reciprocal of atrophy.

7. Which combination correctly matches a muscle type with a UNIQUE regulatory feature, as contrasted in lecture?

Answer: D) Smooth muscle: calmodulin/MLCK, dense bodies

Smooth muscle is regulated by Ca²⁺–calmodulin activating myosin light-chain kinase, anchors actin to dense bodies, and lacks troponin/striations. Skeletal and cardiac use troponin; only skeletal regenerates via satellite cells; cardiac has intercalated discs.

8. A polyanionic, non-sulfated, very large glycosaminoglycan forms the backbone of proteoglycan aggregates and gives ground substance its hydrated-gel property. Enzymatic removal of it would MOST directly:

Answer: C) Reduce water-binding/turgor of ground substance

Hyaluronic acid (non-sulfated GAG) binds enormous amounts of water and anchors proteoglycans, producing the hydrated gel. Removing it collapses the water-holding capacity/turgor of ground substance.

9. Which collagen-type matching is INCORRECT?

Answer: A) Type IV — main fibre of tendon

Tendon is dominated by Type I collagen, not Type IV. Type IV is the mesh of the basal lamina. The other pairings are all correct as taught.

Section 2 · Cartilage, Bone & Integument (Lec 4–6)

10. A 7-year-old presents with multiple low-impact fractures, blue sclerae, dental fragility and conductive hearing loss. Reasoning from the defective protein, which OTHER tissue finding would you most expect?

Answer: C) Weak tendons, lax joints and thin skin

Osteogenesis imperfecta is a Type I collagen defect. Because Type I is also the major fibre of tendon, ligament and dermis, those Type I-rich tissues are likewise weakened (lax joints, thin translucent skin) — the blue sclera itself reflects thin scleral collagen.

11. Articular (hyaline) cartilage of the knee heals poorly after injury. Using the properties taught, which single factor is the MOST important reason?

Answer: B) No perichondrium and avascular (diffusion only)

Articular cartilage is avascular (nutrition by diffusion from synovial fluid) and its joint surface lacks a perichondrium, so there is no local source of progenitor chondroblasts and slow diffusion limits metabolism — hence poor healing.

12. In the epiphyseal growth plate, lengthening of the bone occurs because of mitosis in one zone, while chondrocytes die in another. Match correctly:

Answer: C) Length at proliferation; death at calcification

The proliferation zone (stacked chondrocytes dividing) adds length; in the zone of calcification chondrocytes die as the matrix calcifies, allowing ossification to advance. Reserve is resting cartilage.

13. A bone-collar forms early in long-bone development. According to the lecture sequence, the bone collar is laid down:

Answer: B) Around the diaphysis during primary ossification

In endochondral ossification, a periosteal bone collar forms around the diaphysis as the primary ossification centre develops — it precedes/supports invasion of the cartilage model and is part of primary bone formation.

14. An anti-resorptive hormone used in osteoporosis directly inhibits the bone-resorbing multinucleated cell. Identify the cell and its origin correctly:

Answer: C) Osteocyte — a trapped former osteoblast

Calcitonin inhibits the osteoclast, a large multinucleated cell of haematopoietic/monocyte lineage that resorbs bone at its ruffled border. Osteoblasts (mesenchymal) build bone and become osteocytes.

15. A hypodermic needle is pushed perpendicularly into the skin of the forearm. Which is the correct order of the FIRST FOUR structures pierced?

Answer: D) Epidermis → papillary dermis → reticular dermis → hypodermis

Surface inward: epidermis, then papillary (superficial) dermis, then reticular (deep) dermis, then hypodermis (the actual target of a 'hypodermic' injection).

16. Meissner corpuscles (fine-touch) and Pacinian corpuscles (pressure/vibration) lie at different depths. A superficial second-degree burn that abolishes fine touch but spares deep pressure sense has most damaged receptors located in the:

Answer: E) The dermal papillae (superficial dermis)

Meissner corpuscles sit superficially in the dermal papillae (fine touch); Pacinian corpuscles lie deep in the reticular dermis/hypodermis (pressure/vibration). A superficial burn destroys the superficial Meissner receptors first.

17. In the stratum corneum, the protease-inhibitor LEKTI normally restrains KLK proteases that break down corneodesmosomes. With LEKTI loss-of-function, the predicted consequence is:

Answer: B) Excess KLK degrades corneodesmosomes early

LEKTI inhibits kallikreins (KLK). Without it, KLK over-activity degrades corneodesmosomes too early, weakening the barrier and causing excessive/premature desquamation.

18. Which statement about cartilage taught in lecture is TRUE?

Answer: C) Discs and menisci are fibrocartilage (hyaline component)

Articular discs and menisci are fibrocartilage (with a hyaline component as past papers note). Fibrocartilage lacks a true perichondrium, its main collagen is Type I, and cartilage grows by appositional/interstitial growth — not intramembranous ossification.

19. Albinism, as described, results from:

Answer: C) Lack of tyrosinase, so no melanin

In albinism, melanocytes are present but cannot make melanin because tyrosinase (which converts tyrosine → melanin) is deficient/absent. This is distinct from vitiligo (loss of melanocytes).

Section 3 · Back, Suboccipital Triangle & Vertebral Column (Lec 7–9)

20. A stab wound in the lower back enters through the lumbar (Petit) triangle. From the boundaries taught, which structures form its borders?

Answer: E) Latissimus dorsi, external oblique, iliac crest

The lumbar triangle of Petit is bounded by latissimus dorsi (medially), external abdominal oblique (laterally) and the iliac crest (inferiorly) — a site of lumbar hernia / abscess emergence.

21. Breath sounds are auscultated over the triangle of auscultation. Its boundaries are:

Answer: A) Latissimus dorsi, trapezius, medial scapula

The triangle of auscultation is bounded by trapezius, latissimus dorsi and the medial scapular border (rhomboid major) — where the chest wall is least covered by muscle, so breath sounds are clearest.

22. During neurosurgery in the suboccipital triangle, the surgeon must protect the vessel running across its floor. Identify the contents and the correct superolateral boundary:

Answer: D) Vertebral a. + C1 n.; superolateral oblique sup.

The suboccipital triangle contains the vertebral artery and suboccipital nerve (posterior ramus of C1). Boundaries: rectus capitis posterior major (superomedial), obliquus capitis superior (superolateral), obliquus capitis inferior (inferolateral); roof = semispinalis capitis.

23. A patient with a deep neck laceration loses sensation but retains motor function of trapezius. Recalling trapezius innervation taught, which is correct?

Answer: A) Motor from accessory (XI); sensory from C3–C4

Trapezius receives motor fibres from the spinal accessory nerve (CN XI) and sensory (proprioceptive) fibres from C3–C4. So an isolated sensory deficit can occur while motor (XI) is intact.

24. Which structure carries ONLY motor (efferent) fibres, with NO sensory component?

Answer: D) Ventral (anterior) root

The ventral root is purely motor (efferent). The dorsal root is purely sensory (its ganglion holds sensory cell bodies); the spinal nerve is mixed.

25. A herniated L4–L5 disc compresses a nerve root. Based on the lecture, disc herniations characteristically extend in which direction and most commonly occur at which levels?

Answer: B) Posterolaterally directed; L4–L5 or L5–S1

The annulus fibrosus is thinnest posterolaterally and the posterior longitudinal ligament is narrow there, so herniations extend posterolaterally; ~95% of lumbar disc protrusions are at L4–L5 or L5–S1, producing sciatica.

26. A whiplash (severe neck hyperextension) injury most stretches/tears which ligament, and why?

Answer: C) Posterior longitudinal lig. — resists flexion

The anterior longitudinal ligament prevents hyperextension of the spine; in a whiplash (rear-end collision) the neck hyperextends and this ligament is severely stretched or torn.

27. Rotation of the head ('NO' movement) occurs mainly at which joint, and which structure holds the pivot in place?

Answer: B) Median atlanto-axial pivot; transverse ligament

The median atlanto-axial joint is a pivot where the dens of C2 rotates within the ring of C1; the transverse ligament of the atlas holds the dens, allowing the 'NO' rotation. (The atlanto-occipital joint gives the 'YES' nod.)

28. Rupture of the transverse ligament of the atlas produces which dangerous result?

Answer: D) Freed dens → atlanto-axial subluxation

If the transverse ligament ruptures, the dens is no longer restrained and the median atlanto-axial joint subluxes (incomplete dislocation), endangering the spinal cord at a high cervical level.

29. A patient has an exaggerated lateral curvature of the spine. Recalling the lecture definitions, this is:

Answer: B) Scoliosis

Scoliosis = abnormal lateral curvature. Kyphosis = increased thoracic (posterior) convexity (e.g., from osteoporosis); lordosis = increased lumbar curvature (e.g., pregnancy/potbelly).

30. The intervertebral disc has two parts. The shock-absorbing semifluid core and the surrounding fibrocartilage ring are, respectively:

Answer: E) Nucleus pulposus ; anulus fibrosus

The nucleus pulposus is the semifluid central shock-absorbing core; the anulus fibrosus is the surrounding fibrocartilaginous ring (thinner/less numerous lamellae posteriorly — hence posterolateral herniation).

Section 4 · Scapular, Deltoid, Pectoral & Axilla (Lec 10–11)

31. After axillary lymph-node dissection, a patient cannot adduct/medially rotate the arm well and the lower (sternocostal) part of pectoralis major is weak, but the clavicular head works. Which nerve was injured?

Answer: B) Medial pectoral nerve

The medial pectoral nerve supplies pectoralis minor and the lower (sternocostal) head of pectoralis major; the lateral pectoral nerve supplies the clavicular head. Selective lower-pec weakness localises to the medial pectoral nerve.

32. A patient cannot start abduction of the arm (first ~15°) but, once helped past that, completes abduction normally with the deltoid. Which muscle/nerve is at fault?

Answer: A) Supraspinatus / suprascapular nerve

Supraspinatus initiates the first ~15° of abduction; deltoid (axillary) drives ~15–90°. Failure to initiate with preserved later range localises to supraspinatus (suprascapular nerve).

33. Anterior–inferior dislocation of the glenohumeral joint occurs because the capsule is weakest there. Which nerve and artery are most endangered as they traverse the quadrangular space?

Answer: B) Axillary n. & post. circumflex humeral a.

Anterior–inferior shoulder dislocation (or surgical-neck fracture) threatens the axillary nerve and posterior circumflex humeral artery in the quadrangular space — producing deltoid paralysis and lateral-arm ('regimental badge') sensory loss.

34. A winged scapula (medial border lifts off when pushing against a wall) indicates paralysis of which muscle and nerve?

Answer: D) Serratus anterior / long thoracic n.

Serratus anterior (long thoracic nerve, C5–C7) protracts and stabilises the scapula against the thorax; its paralysis causes medial-border winging, classically tested by pushing against a wall.

35. Which muscle is supplied by the spinal accessory nerve and, when paralysed, causes shoulder droop and weak scapular elevation?

Answer: B) Trapezius

Trapezius is the accessory-nerve (CN XI) muscle of the superficial back; its paralysis drops the shoulder and weakens elevation/retraction of the scapula.

Section 5 · Arm, Forearm & Cubital Fossa (Lec 12–13)

36. A phlebotomist draws blood from the median cubital vein. A protective sheet of tissue separates the vein from the underlying brachial artery and median nerve. That structure, and the correct superficial-to-deep order, are:

Answer: D) Bicipital aponeurosis ; vein→it→artery→nerve

The bicipital aponeurosis (a fascial sheet from the biceps tendon) crosses over and protects the brachial artery and median nerve, separating them from the superficial median cubital vein — which is why cubital venepuncture is relatively safe.

37. A mid-shaft (spiral) humeral fracture causes wrist drop. The nerve and the artery that accompany each other in the spiral groove and may be co-injured are:

Answer: E) Ulnar nerve & sup. ulnar collateral a.

The radial nerve and profunda brachii artery run together in the radial/spiral groove. A mid-shaft spiral fracture injures both — producing wrist drop (extensor loss) and a sensory deficit over the dorsum of the hand.

38. A patient cannot flex the DIP of the index finger or the IP of the thumb and has weak pronation, but sensation is entirely normal. Which nerve is injured and why is sensation spared?

Answer: E) Anterior interosseous nerve — motor only

The anterior interosseous nerve (motor branch of median) supplies flexor pollicis longus, the lateral half of FDP (index) and pronator quadratus. It carries no cutaneous sensory fibres, so the deficit is purely motor (impaired pinch / 'OK sign').

39. A laceration posterior to the medial epicondyle. Using forearm innervation taught, which muscles lose function?

Answer: B) Flexor carpi ulnaris & medial half of FDP

The ulnar nerve passes behind the medial epicondyle and supplies flexor carpi ulnaris and the medial half of FDP (ring + little fingers) in the forearm. These are weakened by a lesion there.

40. A correct forearm muscle–nerve pairing is:

Answer: E) Flexor digitorum superficialis — AIN

Per the slides, the deep branch of the radial nerve supplies ECRB and supinator (before continuing as the posterior interosseous nerve). FPL/FDS are median (AIN for FPL); pronator teres is median; ECU is posterior interosseous.

41. Repetitive wrist flexion in a golfer inflames the common flexor origin. This 'golfer's elbow' involves which structure, and which 'tennis elbow' counterpart is correctly contrasted?

Answer: D) Lateral epicondyle ; medial epicondyle

Golfer's elbow = medial epicondylitis (common flexor origin, repetitive wrist flexion). Tennis elbow = lateral epicondylitis of the common extensor tendon, especially ECRB, from repetitive supination/pronation.

42. A 'Saturday-night palsy' (radial neuropathy) classically follows compression of the radial nerve against the humerus. The expected deficit is:

Answer: A) Claw hand of ring and little fingers

Compression of the radial nerve (e.g., arm draped over a chair) produces wrist drop — loss of wrist and finger extension — because the radial nerve supplies the extensor compartment.

43. A Colles' fracture is described in lecture as:

Answer: D) Distal radius fracture, posterior (dorsal) displacement

Colles' fracture is a distal radius (and often ulna) fracture with posterior/dorsal displacement of the distal fragment ('dinner-fork' deformity), typically from falling onto an outstretched hand.

44. The brachial artery divides into the radial and ulnar arteries. According to the slides, this bifurcation occurs at:

Answer: E) At the apex of the cubital fossa

The brachial artery bifurcates into radial and ulnar arteries at the apex of the cubital fossa, at the level of the neck of the radius.

Section 6 · Hand & Joints of Upper Limb (Lec 14–15)

45. A woman has wrist trauma with weak thumb flexion, abduction, and inability to oppose the thumb, plus wasting of the thenar eminence — but NO sensory loss anywhere. Which structure is injured?

Answer: C) Recurrent (motor) branch of median n.

The recurrent branch of the median nerve supplies the three thenar muscles (abductor pollicis brevis, opponens pollicis, superficial flexor pollicis brevis). Isolated thenar wasting with preserved sensation localises here — the 'ape hand'.

46. Carpal tunnel syndrome compresses the median nerve. Recalling the boundaries taught, the flexor retinaculum (transverse carpal ligament) attaches on the ULNAR side to:

Answer: D) Pisiform bone and hook of hamate

The flexor retinaculum attaches ulnarly to the pisiform and hook of hamate, and radially to the scaphoid tubercle and trapezium ridge, converting the carpal arch into the carpal tunnel that transmits the median nerve and flexor tendons.

47. A cyclist with prolonged handlebar pressure develops weakness of the interossei, adductor pollicis and hypothenar muscles, with a clawed hand but intact thenar opposition. The lesion is:

Answer: A) Deep ulnar branch in Guyon's canal

Guyon-canal compression (handlebar palsy / hook-of-hamate fracture) injures the deep branch of the ulnar nerve, paralysing interossei, adductor pollicis, hypothenar and medial two lumbricals — clawing with first-dorsal-interosseous wasting, while median-supplied thenar opposition is spared.

48. In an ulnar 'claw hand', the MCP joints are hyperextended and the PIP/DIP joints flexed in the 4th–5th digits. The mechanism, per the lecture, is paralysis of which combination?

Answer: D) Lumbricals (medial two) and interossei

Lumbricals and interossei flex MCP and extend the IP joints. With ulnar palsy the medial two lumbricals and all interossei are lost, so the long extensors hyperextend the MCPs and the long flexors flex the IP joints — the claw.

49. Which intrinsic hand muscle is the ODD ONE OUT because it is NOT supplied by the deep branch of the ulnar nerve?

Answer: B) The palmar interossei

Almost all intrinsic hand muscles are deep ulnar; the exceptions are the three thenar muscles (abductor pollicis brevis, opponens pollicis, superficial flexor pollicis brevis) and the lateral two lumbricals — all median. Opponens

pollicis is therefore the exception here.

50. The anatomical snuffbox. From the slides, which set of boundaries/contents is correct?

Answer: A) Medial EPL; lateral EPB/APL; floor scaphoid; radial a.

Snuffbox: medial border = EPL; lateral border = EPB and APL; floor = scaphoid and trapezium; roof contains the superficial radial nerve and cephalic vein origin; content = 2nd part of the radial artery.

51. Which upper-limb joint-type pairing is INCORRECT, per the summary slide?

Answer: B) Proximal radio-ulnar — pivot joint

The wrist (radiocarpal) joint is a condyloid (ellipsoid) joint, not a hinge. SC = saddle, GH = ball-and-socket, elbow = hinge, proximal radio-ulnar = pivot — all correct as taught.

52. A baseball pitcher tears the ulnar collateral ligament of the elbow and undergoes 'Tommy John' reconstruction. The UCL connects which bony points, per the slide?

Answer: B) Medial epicondyle to coronoid and olecranon

The (triangular) ulnar collateral ligament runs from the medial epicondyle to the coronoid process and olecranon of the ulna. Overhead throwing stresses it; rupture is repaired by graft (Tommy John surgery).

53. In a fall on the shoulder tip causing AC joint separation, which ligament — which normally prevents the acromion moving inferiorly when forced medially — is commonly torn along with the AC ligament?

Answer: D) Coracoclavicular ligament tear

The coracoclavicular ligament (conoid + trapezoid parts) suspends the scapula from the clavicle and resists inferior displacement of the acromion. AC separations tear the AC and, in higher grades, the coracoclavicular ligament.

54. 'Nursemaid's elbow' (radial-head subluxation) in a toddler involves which ligament, per the slides?

Answer: C) Anular ligament of the radius

The anular ligament encircles the radial head at the proximal radio-ulnar (pivot) joint; in young children axial traction lets the radial head slip from under the loose anular ligament — nursemaid's elbow.

Section 7 · Development of Bone, Skin & Muscle (Lec 16–17)

55. A neonate has a long, narrow, boat-shaped skull. Reasoning from the suture involved, which suture fused prematurely and what is the term?

Answer: A) Sagittal suture — scaphocephaly

Premature fusion of the sagittal suture restricts width and forces antero-posterior growth, giving the long narrow boat-shaped skull = scaphocephaly (a craniosynostosis).

56. Which skull bone is a CARTILAGINOUS-neurocranium (chondrocranium) derivative, forming by endochondral ossification?

Answer: C) Sphenoid

The chondrocranium/skull base (sphenoid, ethmoid, petrous temporal, occipital) ossifies endochondrally. The flat vault bones (frontal, parietal, squamous temporal) and facial bones (maxilla) are membranous.

57. A child has unilateral absence of the sternocostal head of pectoralis major with ipsilateral hand hypoplasia/syndactyly. This is:

Answer: D) Poland syndrome

Poland syndrome = congenital absence/hypoplasia of pectoralis major (sternocostal head), often with ipsilateral limb anomalies (syndactyly). Prune-belly = absent abdominal-wall musculature.

58. Vertebrae and ribs develop from which somite derivative?

Answer: A) Sclerotome

The sclerotome forms the axial skeleton — vertebrae and ribs. Myotome → skeletal muscle; dermatome → dermis of the back.

59. Fusion of adjacent digits (failure of separation) is termed:

Answer: D) Syndactyly

Syndactyly = webbing/fusion of adjacent digits. Polydactyly = extra digits; brachydactyly = short digits.

60. Partial or complete congenital absence of the abdominal wall musculature characterises:

Answer: A) Prune-belly syndrome

Prune-belly (Eagle-Barrett) syndrome features deficient/absent abdominal-wall muscle, giving the wrinkled 'prune' appearance.

Section 8 · Gluteal Region, Posterior Thigh & Popliteal Fossa (Lec 18–19)

61. A nurse must give a deep IM gluteal injection avoiding the sciatic nerve. Reasoning from the nerve's course, which quadrant is safe and why?

Answer: A) Superolateral — nerve runs lower/medial

The sciatic nerve descends through the inferomedial gluteal region (below piriformis). The superolateral (upper-outer) quadrant is therefore the safe IM injection site.

62. A posterior hip dislocation injures the sciatic nerve, producing foot drop. Which division is most vulnerable, and what is lost?

Answer: E) Common fibular div. — dorsiflexion/eversion

The common fibular (peroneal) division of the sciatic nerve is most vulnerable in posterior hip dislocation; its loss abolishes dorsiflexion and eversion, giving foot drop. The tibial division is relatively spared.

63. A bullet enters the popliteal fossa. Recalling the superficial-to-deep order of the central structures, which is the MOST superficial and which the DEEPEST?

Answer: E) Tibial nerve superficial; artery deepest

From superficial to deep in the fossa: tibial nerve → popliteal vein → popliteal artery (deepest, against the joint capsule/femur). So the nerve is most superficial and the artery deepest.

64. Avascular necrosis of the femoral head most often follows damage to which artery, per the lecture?

Answer: E) Profunda femoris artery

The medial circumflex femoral artery (via retinacular branches) supplies most of the femoral head. Femoral-neck fracture or dislocation disrupting it causes avascular necrosis; the artery of the ligamentum teres contributes little in adults.

65. Which muscle is NOT a 'true hamstring' because it is supplied by the common fibular division of the sciatic nerve rather than the tibial division?

Answer: E) Long head of biceps femoris

The short head of biceps femoris is supplied by the common fibular division, so it is not a 'true' (tibial-division) hamstring. Semitendinosus, semimembranosus and the long head of biceps are tibial-division hamstrings.

66. Which structure passes through the greater sciatic foramen ABOVE piriformis?

Answer: C) Inferior gluteal nerve

The superior gluteal nerve (and vessels) exit above piriformis; almost everything else (sciatic, inferior gluteal, pudendal, posterior cutaneous nerve of thigh) exits below piriformis.

67. Gluteus medius and minimus share an action important for gait. Their paralysis (e.g., superior gluteal nerve injury) produces which sign?

Answer: B) Trendelenburg sign (pelvis drop)

Gluteus medius/minimus (superior gluteal nerve) abduct and medially rotate the thigh and stabilise the pelvis during single-leg stance. Their weakness causes a positive Trendelenburg sign — the unsupported side of the pelvis drops.

68. Gluteus maximus, the chief extensor and lateral rotator of the thigh, is innervated by the:

Answer: B) Inferior gluteal nerve

Gluteus maximus is supplied by the inferior gluteal nerve and extends and laterally rotates the thigh (important for rising from sitting and climbing).

Section 9 · Anterior/Medial Thigh, Leg, Foot & Lower-limb Joints (Lec 20–23)

69. To begin flexion of the fully extended, 'locked' (screw-home) knee, one muscle unlocks it by laterally rotating the femur on the fixed tibia. Which one, and what happens if it is torn?

Answer: D) Popliteus — knee stays locked in extension

Popliteus unlocks the extended knee by laterally rotating the femur on a fixed tibia (or medially rotating the tibia), reversing the screw-home mechanism. Its dysfunction impairs the ability to initiate knee flexion.

70. A fracture at the neck of the fibula injures the most commonly damaged nerve of the lower limb. Identify the nerve and the full deficit:

Answer: E) Common fibular n. — foot drop (dorsi+evert)

The common fibular nerve winds subcutaneously around the fibular neck (vulnerable). Loss of its deep branch (dorsiflexors) and superficial branch (evertors) gives foot drop with absent dorsiflexion and eversion.

71. At heel-strike (initial contact) of the gait cycle, the correct joint positions are:

Answer: C) Hip flex, knee extend, ankle dorsiflex

At heel-strike the hip is flexed, the knee extended to accept load, and the ankle dorsiflexed as the heel contacts; the anterior-compartment dorsiflexors act eccentrically to lower the foot.

72. Obturator-nerve compression in the obturator canal weakens which movement, and which sensory area is affected?

Answer: A) Hip adduction; medial-thigh skin

The obturator nerve supplies the medial (adductor) compartment — adductor longus/brevis/magnus (adductor part), gracilis, obturator externus — and skin of the medial thigh. Compression weakens hip adduction with medial-thigh numbness.

73. Collapse of the medial longitudinal arch with hindfoot eversion/abduction is which deformity, and dysfunction of which tendon (the dynamic arch support) is a classic adult cause?

Answer: C) Pes planus — tibialis posterior dysfunction

Flatfoot (pes planus) is collapse of the medial longitudinal arch with eversion/abduction; tibialis posterior is the chief dynamic supporter of that arch and its tendon dysfunction is a leading acquired adult cause.

74. The anterior compartment of the leg is supplied by which nerve and artery, and its main action is:

Answer: D) Deep fibular n., ant. tibial a.; dorsiflex

The anterior leg compartment (tibialis anterior, EDL, EHL, fibularis tertius) is supplied by the deep fibular nerve and anterior tibial artery and produces dorsiflexion (and toe extension).

75. The lateral compartment of the leg (fibularis longus and brevis) is supplied by which nerve and produces which movement?

Answer: E) Superficial fibular n.; eversion

The lateral compartment (fibularis/peroneus longus and brevis) is supplied by the superficial fibular nerve and everts the foot (with some plantarflexion).

76. Tarsal-tunnel syndrome compresses a nerve behind the medial malleolus, deep to the flexor retinaculum. Identify the nerve and a resulting deficit:

Answer: A) Sural nerve; lateral-foot numbness

The tibial nerve passes through the tarsal tunnel (behind the medial malleolus, deep to the flexor retinaculum) before dividing into medial and lateral plantar nerves. Compression gives sole numbness and intrinsic foot-muscle weakness.

77. The deep posterior compartment of the leg contains popliteus, FDL, FHL and tibialis posterior, all supplied by the tibial nerve. Tibialis posterior's main action is:

Answer: D) Plantarflexion with inversion (arch)

Tibialis posterior plantarflexes and inverts the foot and is the key dynamic supporter of the medial longitudinal arch; it lies in the deep posterior compartment supplied by the tibial nerve.

78. The hip joint is a multiaxial ball-and-socket joint. Which ligament is the strongest, limits hyperextension, and has a Y-shape attaching to the AIIS?

Answer: D) Ischiofemoral ligament

The iliofemoral ligament is the strongest hip ligament (Y-shaped, attaching to the AIIS/acetabular rim and the intertrochanteric line). It limits hyperextension and helps maintain upright posture.

79. A 'posterior drawer' sign (tibia displaces posteriorly on the fixed femur) indicates injury to which ligament, and where does it attach on the tibia?

Answer: B) PCL — posterior intercondylar area

The posterior cruciate ligament arises from the posterior intercondylar area of the tibia and prevents posterior tibial displacement; a positive posterior drawer indicates PCL injury. (The ACL arises anteriorly and prevents anterior translation.)

80. Which knee meniscus is C-shaped, broader posteriorly, and firmly adherent to the deep surface of the tibial collateral ligament — making it more often injured?

Answer: A) The medial meniscus

The medial meniscus is C-shaped and firmly attached to the tibial (medial) collateral ligament, so combined valgus injuries tend to tear it. The lateral meniscus is more O-shaped and more mobile.

81. The acetabulum is formed by the fusion of which three bones at the triradiate cartilage?

Answer: A) Ilium, ischium, pubis

The acetabulum is formed by ilium, ischium and pubis meeting at the triradiate cartilage; the sacrum is not part of it.

82. The great saphenous vein, per the slides, ends by emptying into the:

Answer: E) The femoral vein (saphenous opening)

The great saphenous vein ascends the medial limb, traverses the saphenous opening in the fascia lata, and drains into the femoral vein. (The small saphenous vein empties into the popliteal vein.)

83. The ankle (talocrural) joint is most stable and 'close-packed' in which position, given the talus geometry taught?

Answer: D) Dorsiflexion (wide talus wedged in)

The trochlea of the talus is wider anteriorly; in dorsiflexion the broad anterior talus wedges into the mortise, making the joint most stable. Plantarflexion brings the narrow posterior part into the mortise — less stable, more prone to sprain (typically inversion in plantarflexion).

Section 10 · Integrative Histology, Cartilage & Bone — Challenge Round

84. A researcher stains a trachea section. The cartilage shows chondrocytes in lacunae, a basophilic territorial matrix, and a perichondrium; collagen is not visible by routine H&E because its refractive index matches the matrix. The cartilage type and its 'hidden' collagen are:

Answer: D) Hyaline cartilage; Type II (masked)

Tracheal rings are hyaline cartilage; its Type II collagen fibrils have the same refractive index as the ground substance, so they are 'masked' (not seen) on routine H&E. Basophilia comes from sulfated proteoglycans of the territorial matrix.

85. Two students argue about appositional vs interstitial cartilage growth. Which statement is correct?

Answer: D) Appositional: chondrocytes divide in lacunae

Interstitial growth = chondrocytes divide within the matrix forming isogenous (cell-nest) groups, expanding cartilage from within. Appositional growth = chondroprogenitors at the perichondrium add new cartilage at the surface.

86. During fracture repair, the stages must occur in order. Which sequence is correct?

Answer: B) Haematoma → soft callus → bony callus → remodel

Fracture healing: (1) haematoma, (2) fibrocartilaginous/soft callus, (3) bony/hard callus (osteoblasts lay osteoid), (4) bone remodelling. Osteoblasts first make osteoid in the bony-callus stage.

87. Pathologic loss of bone mass that reduces strength and predisposes to fracture, distinct from defective mineralisation, is:

Answer: E) Osteoporosis

Osteoporosis = reduced bone MASS (matrix + mineral) compromising strength. Osteomalacia is defective MINERALISATION of osteoid. Distinguishing 'mass' vs 'mineralisation' is a common trap.

88. Which feature does NOT describe an osteoclast?

Answer: A) Resorbs bone at a ruffled border

Osteoclasts resorb bone; they do NOT secrete osteoid — that is the osteoblast's job. Osteoclasts are large, multinucleated, monocyte-derived, and inhibited by calcitonin.

89. Which statement about osteoblasts is FALSE?

Answer: C) They secrete alkaline phosphatase

Osteoblasts FORM bone (osteoid, alkaline phosphatase) and become osteocytes when surrounded by matrix; they do NOT resorb bone with a ruffled border (that is the osteoclast).

Section 11 · Upper-limb Nerve Lesions — Differentiation Round

90. Four hand deformities are taught: ape hand, claw hand, hand of benediction, wrist drop. Match each to its lesion. Which pairing is CORRECT?

Answer: A) Benediction (on fist) — median at elbow

Median nerve injury at the elbow gives the 'hand of benediction' when making a fist (index/middle can't flex). Ape hand = recurrent median (thenar loss); claw hand = ulnar; wrist drop = radial.

91. A patient has a claw hand that is MORE pronounced (deeper clawing of ring/little fingers) when the ulnar lesion is at the WRIST than at the elbow. The reason ('ulnar paradox') taught is:

Answer: C) Wrist lesion spares FDP, so DIPs still flex

In the 'ulnar paradox', a distal (wrist) lesion spares the ulnar-supplied medial FDP, so the DIP flexors of the 4th/5th fingers still work and pull those fingers into deeper flexion — the claw looks worse than with a proximal (elbow) lesion that also paralyses FDP.

92. A surgical-neck-of-humerus fracture causes which deficit, and which test confirms it?

Answer: D) Deltoid palsy + badge numbness; test abduction

The surgical neck of the humerus is crossed by the axillary nerve (and posterior circumflex humeral artery). Fracture there paralyses the deltoid (weak abduction 15–90°) and numbs the 'regimental badge' area over the lateral arm.

93. Which cutaneous nerve of the upper limb is NOT derived from the posterior cord?

Answer: A) Post. cutaneous n. of forearm

The lateral cutaneous nerve of the forearm is the continuation of the musculocutaneous nerve (lateral cord), NOT the posterior cord. The other listed cutaneous nerves arise from posterior-cord branches (axillary, radial).

94. For surgery at the SURGICAL NECK of the humerus, which brachial-plexus block is preferred, per the past-paper teaching?

Answer: E) Interscalene block

An interscalene block (at the roots/trunks) covers the shoulder/proximal humerus well and is preferred for surgery around the surgical neck/shoulder. Axillary blocks are better for forearm/hand surgery.

95. Klumpke palsy (lower-trunk C8–T1) classically produces which hand sign?

Answer: A) Erb 'waiter's tip' posture

Klumpke palsy injures the lower trunk (C8–T1), affecting intrinsic hand muscles and producing an ulnar-type claw hand (and possible Horner's). Erb palsy (upper trunk, C5–C6) gives the 'waiter's tip'.

96. Pain on resisted elbow flexion when the forearm is SUPINATED points most to injury of which muscle?

Answer: D) Coracobrachialis

Biceps brachii is the strong flexor/supinator of the forearm; resisted flexion in supination loads it maximally, so pain there localises to biceps. Brachioradialis is tested in mid-pronation; brachialis flexes in all positions but is deep.

Section 12 · Forearm & Hand Detail — Precision Round

97. The superficial flexor muscles of the forearm share a common origin. Which muscle is the ODD ONE because it is supplied by the ULNAR (not median) nerve?

Answer: D) Palmaris longus

All superficial flexors arise from the common flexor tendon (medial epicondyle); pronator teres, FCR, palmaris longus and FDS are median, but flexor carpi ulnaris is supplied by the ulnar nerve — the exception.

98. Which deep-layer forearm flexor has a SPLIT nerve supply, with its medial part ulnar and lateral part anterior interosseous (median)?

Answer: E) Pronator quadratus

Flexor digitorum profundus has dual innervation: the medial part (ring/little) by the ulnar nerve, the lateral part (index/middle) by the anterior interosseous nerve (median). This split underlies the ulnar paradox.

99. The 'outcropping' muscles of the thumb (APL, EPB, EPL) emerge from the forearm and are all supplied by which nerve?

Answer: A) Anterior interosseous nerve

The outcropping thumb muscles — abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus — are deep extensors supplied by the posterior interosseous nerve (continuation of the deep radial nerve).

100. Six osseofibrous tunnels under the extensor retinaculum transmit 12 tendons of nine extensor muscles on the dorsum of the wrist. The retinaculum spans between which two bones?

Answer: E) The ulna and radius

The extensor retinaculum attaches to the ulna and radius, creating six osseofibrous tunnels for the synovial sheaths of the extensor tendons on the dorsum of the hand.

101. The lumbricals are unusual muscles. According to the slides, they ORIGINATE from and produce which action?

Answer: E) From FDP tendons; flex MCP, extend IP

Lumbricals arise from the FDP tendons and insert into the extensor expansions; they flex the MCP joints and extend the IP joints (digits 2–5). Lateral two are median, medial two are deep ulnar.

102. Dorsal interossei ABduct and palmar interossei ADduct the fingers relative to the axial line. A useful mnemonic mapping is 'PAD/DAB'. The axial reference line passes through which digit?

Answer: C) Middle (3rd) finger

The axial line passes through the middle (3rd) finger. Dorsal interossei ABduct fingers away from it (DAB), palmar interossei ADduct toward it (PAD). All are deep-ulnar supplied.

103. The deep palmar arch is formed mainly by which artery, and gives rise to which named branches?

Answer: B) Radial a. (3rd part); princeps + radialis indicis

The deep palmar arch is the continuation of the radial artery (3rd part) anastomosing with the deep branch of the ulnar artery; it gives the princeps pollicis and radialis indicis and palmar metacarpal arteries.

Section 13 · Joints — Classification & Movement Round

104. Match each joint to its type. Which mapping is WRONG?

Answer: E) Interphalangeal — hinge joint

The proximal radio-ulnar joint is a PIVOT joint (radial head rotating in the annular ligament/radial notch), not plane. The others are correct: SC and thumb CMC = saddle; MCP = condyloid; IP = hinge.

105. Movements of the sternoclavicular joint, the only joint linking the upper limb to the axial skeleton, include all EXCEPT:

Answer: B) Posterior rotation of the girdle

The SC joint allows elevation/depression, protraction/retraction and (some) posterior rotation of the pectoral girdle. Pronation/supination is a forearm (radio-ulnar) movement, not an SC-girdle motion.

106. The glenohumeral joint sacrifices stability for mobility. Which structure deepens the shallow glenoid cavity?

Answer: B) Glenoid labrum (fibrocartilage)

The fibrocartilaginous glenoid labrum deepens the glenoid cavity to improve congruence and stability of the ball-and-socket shoulder joint, while the rotator cuff provides dynamic support.

107. In supination, which two muscles are the prime movers, per the radio-ulnar joint slide?

Answer: C) Supinator and biceps brachii

Supination is produced by the supinator and biceps brachii (biceps is a powerful supinator when the elbow is flexed). Pronation is by pronator teres and pronator quadratus.

108. The subacromial (subdeltoid) bursa lies between which structures, and its inflammation (bursitis) causes what?

Answer: D) Acromion/deltoid and supraspinatus; arc pain

The subacromial/subdeltoid bursa sits between the acromion, coraco-acromial ligament and deltoid (above) and the supraspinatus tendon (below). Bursitis there causes painful, restricted abduction (impingement-type pain).

Section 14 · Lower-limb Compartments, Vessels & Gait — Applied Round

109. A young man has an anterior-compartment-of-leg crush injury with rising pressure. Reasoning from its contents, the EARLIEST sign of anterior compartment syndrome and the nerve at risk are:

Answer: A) Pain on stretch + web-space numb; deep fibular

The anterior compartment holds the deep fibular nerve. Compartment syndrome causes pain on passive stretch (passive toe flexion stretches the dorsiflexors) and early sensory loss in the first web space (deep fibular territory), before pulses disappear.

110. The popliteal artery is the deepest central structure of the popliteal fossa. It continues as which arteries, per the lecture?

Answer: B) Anterior tibial and posterior tibial

The popliteal artery gives genicular branches around the knee and then divides into the anterior tibial and posterior tibial arteries; the posterior tibial gives off the fibular (peroneal) artery.

111. During the stance-to-push-off transition of gait, the triceps surae (gastrocnemius + soleus) act to:

Answer: A) Plantarflex for push-off

The triceps surae (gastrocnemius and soleus, via the calcaneal tendon) plantarflex the ankle, providing the push-off ('toe-off') power of gait. Loss gives a weak, ineffective push-off.

112. The small saphenous vein and the great saphenous vein terminate, respectively, in the:

Answer: B) Popliteal vein ; femoral vein

The small saphenous vein drains into the popliteal vein; the great saphenous vein drains into the femoral vein (through the saphenous opening). Don't swap them.

113. Foot drop from common fibular nerve injury produces a characteristic gait. Which description fits, per the lecture?

Answer: B) Steppage gait with foot-slap

Common fibular nerve palsy paralyses the dorsiflexors (anterior compartment) and evertors (lateral compartment), so the foot drops; the patient lifts the knee high (steppage) and the foot slaps down — the classic foot-drop gait.

114. The medial circumflex femoral artery is critical for the femoral head. Which other arteries contribute to the hip joint's blood supply, per the joints lecture?

Answer: D) Retinacular, lig. teres, circumflex aa.

The hip joint is supplied by the retinacular arteries, the artery to the head of the femur (in the ligamentum teres), and the medial and lateral circumflex femoral arteries — the medial circumflex being the dominant supply to the head.

Section 15 · Spine, Development & Integrative — Final Challenge

115. A weightlifter develops lumbar canal stenosis with neurogenic claudication. Hypertrophy/buckling of which posterior-wall ligament most narrows the canal?

Answer: C) Posterior longitudinal lig.

The ligamentum flavum lines the posterior wall of the vertebral canal; age/stress-related hypertrophy and buckling are a leading cause of acquired lumbar central stenosis and neurogenic claudication.

116. The primary curvatures of the spine (present in the fetus) versus the secondary curvatures (developing after birth) are correctly listed as:

Answer: C) Primary thoracic/sacral; secondary cervical/lumbar

Thoracic and sacral kyphoses are PRIMARY (present in fetal flexed posture). Cervical lordosis (head-raising) and lumbar lordosis (standing/walking) are SECONDARY curvatures that develop after birth.

117. The dens (odontoid) of C2 is held against the anterior arch of C1 by a ligament stronger than the dens itself. That ligament and the joint type are:

Answer: E) Transverse ligament; median pivot joint

The transverse ligament of the atlas holds the dens at the median atlanto-axial PIVOT joint (head rotation). It is stronger than the dens, so trauma tends to fracture the dens rather than rupture the ligament — though ligament rupture frees the dens (subluxation).

118. Which craniovertebral joint provides the 'YES' nod (flexion/extension), and which provides the 'NO' rotation?

Answer: C) Atlanto-occipital YES; atlanto-axial NO

Atlanto-occipital joint = flexion/extension ('YES' nod). Median atlanto-axial joint = rotation ('NO' shake), pivoting on the dens. A classic past-paper pairing.

119. Reasoning across systems: a patient with osteoporosis develops an exaggerated thoracic kyphosis ('dowager's hump'). The structural cause taught is:

Answer: A) Anterior wedge fractures of thoracic bodies

Osteoporotic thoracic vertebral bodies undergo anterior wedge compression fractures, increasing the normal thoracic kyphosis into the abnormal 'dowager's hump'.

120. Spondylolisthesis, contrasted with spondylolysis and spondylitis, is best described as:

Answer: D) Forward slippage of one vertebral body on the one below

Spondylolisthesis = forward slippage of a vertebra on the one below (often following bilateral pars defects).

Spondylolysis = pars defect without slip; spondylitis = vertebral inflammation.

121. A child has bilateral clubfoot (talipes equinovarus). The deformity, per the development lecture, fundamentally involves abnormal position/development of which bone?

Answer: D) The talus (foot inverted/plantarflexed)

Talipes equinovarus (clubfoot) centres on a deformity involving the talus, with the hindfoot in equinus/varus and the forefoot adducted — the foot is plantarflexed and inverted.

122. Hypertrichosis is described as resulting from abnormal development of the:

Answer: E) Hair follicle

Hypertrichosis (excessive hair) arises from abnormal development/activity of hair follicles.

123. Premature bilateral fusion of the CORONAL sutures produces a short, wide, tall skull termed:

Answer: D) Brachycephaly

Bilateral coronal synostosis limits antero-posterior growth, producing a short/broad (often tall) skull = brachycephaly. (Sagittal = scaphocephaly; metopic = trigonocephaly; unilateral = plagiocephaly.)

124. A femoral hernia, contrasted with an inguinal hernia in the past papers, protrudes:

Answer: C) Lateral to tubercle, above ligament

A femoral hernia passes through the femoral canal, emerging below the inguinal ligament and lateral/inferior to the pubic tubercle — distinguishing it from an inguinal hernia, which is above the ligament and medial to the tubercle.

Section 16 · Mixed Clinical Vignettes — Capstone Round

125. A 22-year-old falls on an outstretched hand and has tenderness in the anatomical snuffbox. Reasoning from the snuffbox floor and its blood supply, which bone is likely fractured and what complication is feared?

Answer: C) Scaphoid; avascular necrosis risk

The snuffbox floor includes the scaphoid; snuffbox tenderness after a fall suggests scaphoid fracture. Its blood supply enters distally (retrograde), so a waist/proximal fracture risks avascular necrosis of the proximal pole.

126. A diabetic has burning sole pain and tingling worse at night, relieved by ankle movement, with a positive tap sign behind the medial malleolus. The compressed nerve and tunnel are:

Answer: E) Tibial nerve; the tarsal tunnel

Tarsal-tunnel syndrome compresses the tibial nerve behind the medial malleolus, deep to the flexor retinaculum. Sole burning/tingling and a positive Tinel (tap) sign there are characteristic.

127. A footballer is struck on the LATERAL knee (valgus force). Per the 'unhappy triad' logic and the meniscal attachments taught, which combination is most likely injured together?

Answer: E) Tibial collateral, medial meniscus, ACL

A valgus blow tensions the medial side: it tears the tibial (medial) collateral ligament, which is adherent to the medial meniscus (so that tears too), and commonly the ACL — the classic 'unhappy triad'.

128. A patient cannot extend the knee and has lost the patellar reflex, with anterior-thigh sensory loss. The nerve and compartment are:

Answer: D) Femoral nerve; anterior compartment

The femoral nerve supplies the anterior thigh (quadriceps → knee extension, patellar reflex) and gives the saphenous nerve (medial leg) plus anterior-thigh cutaneous branches. Its injury abolishes knee extension and the knee jerk.

129. After prolonged lithotomy positioning, a patient has weak hip adduction and numb medial thigh, with intact knee extension. The nerve is:

Answer: E) Obturator nerve

The obturator nerve (medial/adductor compartment) can be compressed in the obturator canal or by positioning; it weakens hip adduction and numbs the medial thigh while sparing femoral-supplied knee extension.

130. A swimmer with shoulder pain on overhead motion has a painful arc around 60–120° of abduction and weakness initiating abduction. The most likely structure and bursa involved are:

Answer: B) Supraspinatus tendon; subacromial bursa

Supraspinatus runs under the acromion; its tendon and the subacromial/subdeltoid bursa are compressed during mid-range abduction, giving a painful arc (~60–120°) and weak initiation of abduction — subacromial impingement.

131. A cyclist leaning on the handlebars develops wasting of the first dorsal interosseous and cannot cross the fingers, but thumb opposition and sensation over the thenar area are normal. Localise precisely:

Answer: E) Deep ulnar branch in Guyon's canal

A purely motor deficit of ulnar intrinsics (interossei, adductor pollicis, hypothenar) with NO sensory loss and SPARED proximal FDP/FCU localises to the deep (motor) branch of the ulnar nerve in Guyon's canal — handlebar palsy.

132. A toddler is lifted by one arm and then refuses to use it, holding the forearm pronated and slightly flexed. The diagnosis and the structure involved are:

Answer: E) Radial-head subluxation; annular ligament

Axial traction on a young child's extended, pronated arm subluxes the radial head from under the loose annular ligament — nursemaid's elbow; the child holds the arm pronated and resists supination.

133. A patient with long-standing carpal tunnel syndrome has thenar wasting and cannot oppose the thumb ('ape hand'). Yet sensation over the central palm is relatively preserved. The anatomical reason for spared central-palm sensation is:

Answer: E) Palmar cutaneous branch bypasses the tunnel

The palmar cutaneous branch of the median nerve arises proximal to the flexor retinaculum and runs superficial to the tunnel, so its territory (central palm) is spared even when the median nerve is compressed within the carpal tunnel.

134. Reasoning from joint type: a 'pivot' synovial joint permits which single movement, and which lower-/upper-limb examples fit?

Answer: D) Rotation; radio-ulnar and atlanto-axial

A pivot joint permits rotation about a central axis. Examples: proximal and distal radio-ulnar joints (pronation/supination) and the median atlanto-axial joint (head rotation).

135. A marathon runner develops medial-arch pain and visible flattening of the arch when standing. Which structure, the dynamic supporter taught, has likely failed, and which passive structure also supports the arch?

Answer: A) Tibialis posterior; plantar aponeurosis/spring

Tibialis posterior is the chief DYNAMIC supporter of the medial longitudinal arch; its dysfunction causes acquired flatfoot. Passive support comes from the plantar aponeurosis and the spring (plantar calcaneonavicular) ligament.

136. A boxer fractures the neck of the 5th metacarpal ('boxer's fracture'). Which intrinsic muscle group, attaching to that region and supplied by the deep ulnar nerve, may be affected in grip?

Answer: E) The hypothenar muscle group

The hypothenar muscles (abductor digiti minimi, flexor digiti minimi brevis, opponens digiti minimi) act on the 5th digit/metacarpal and are deep-ulnar supplied; a 5th-metacarpal injury can affect little-finger grip mechanics.

137. Integrative: a patient with a high radial-nerve lesion (axilla) has wrist drop AND loss of elbow extension, whereas a mid-shaft (spiral groove) lesion spares elbow extension. The reason is:

Answer: E) Triceps branches arise above the groove

The radial nerve gives its triceps (elbow-extension) branches high in the arm, proximal to the spiral groove. A spiral-groove (mid-shaft) lesion therefore spares triceps (elbow extension preserved) but still causes wrist drop; a higher axillary lesion knocks out triceps too.

138. A patient cannot abduct the arm beyond ~15° and the deltoid is wasted with lateral-arm numbness, after a blow to the shoulder. Trace the lesion: which nerve, from which cord, through which space?

Answer: C) Axillary n.; post. cord; quadrangular space

The axillary nerve (from the posterior cord, C5–C6) passes through the quadrangular space with the posterior circumflex humeral artery to supply deltoid and teres minor and the skin over the lower deltoid. Injury gives weak abduction (>15°) and 'badge' numbness.

139. Final integrative trap: which statement correctly distinguishes the action loss in a TIBIAL nerve injury from a COMMON FIBULAR nerve injury at the knee?

Answer: E) Tibial → plantarflex/invert loss; fibular → foot drop

Tibial nerve injury abolishes plantarflexion and inversion and sole sensation (calcaneal/plantar branches). Common fibular injury abolishes dorsiflexion (deep branch) and eversion (superficial branch) → foot drop. Mixing these is a classic exam error.

Section 17 · Rapid High-Yield Round

140. Carpal-tunnel syndrome compresses which nerve, and which thumb function is classically lost?

Answer: C) Median nerve; opposition of the thumb

Carpal-tunnel syndrome compresses the median nerve under the flexor retinaculum; thenar weakness causes loss of thumb opposition ('ape hand') with lateral 3½-digit sensory changes.

141. Which rotator-cuff muscle is NOT one of the four (SITS) listed in the glenohumeral slide?

Answer: D) Teres major

The rotator cuff is supraspinatus, infraspinatus, teres minor and subscapularis (SITS). Teres major is NOT part of the cuff.

142. The chief flexor of the elbow joint and the chief extensor, per the elbow-joint slide, are respectively:

Answer: D) Brachialis; triceps brachii

The slide names brachialis as the chief flexor of the elbow (flexes in all positions) and triceps brachii as the chief extensor.

143. Which compartment of the leg, if its nerve is injured, abolishes EVERSION while sparing dorsiflexion?

Answer: C) Lateral compartment (superficial fibular)

Isolated superficial fibular nerve injury (lateral compartment) abolishes eversion (fibularis longus/brevis) but spares dorsiflexion (deep fibular, anterior compartment).

144. The erector spinae, chief extensor of the vertebral column, splits into three columns. From lateral to medial they are:

Answer: B) Iliocostalis, longissimus, spinalis

Erector spinae columns lateral→medial: Iliocostalis (lateral), Longissimus (intermediate), Spinalis (medial) — mnemonic 'I Like Standing'.

145. The transversospinalis group, deep to erector spinae, from superficial to deep is:

Answer: E) Semispinalis, multifidus, rotatores

Transversospinalis superficial→deep: semispinalis (spans 4–6 segments), multifidus (2–4), rotatores (1–2). They run from transverse processes to spinous processes of more superior vertebrae.

146. Which gluteal-region muscle, paralysis of which gives a positive Trendelenburg sign, is supplied by the superior gluteal nerve?

Answer: B) Gluteus medius (and minimus)

Gluteus medius/minimus (superior gluteal nerve) stabilise the pelvis in single-leg stance; their weakness causes the unsupported pelvis to drop — positive Trendelenburg.

147. The thenar muscles (abductor pollicis brevis, opponens pollicis, superficial flexor pollicis brevis) are supplied by which nerve?

Answer: E) Recurrent branch of the median nerve

The three thenar muscles are supplied by the recurrent (motor) branch of the median nerve — the exception to the deep-ulnar supply of most intrinsic.

148. The interosseous membrane between radius and ulna is which kind of fibrous joint?

Answer: E) Syndesmosis

The radio-ulnar interosseous membrane is a syndesmosis (a fibrous joint uniting bones by a sheet/ligament).

149. Which intervertebral ligament limits flexion and is elastic, helping straighten the column after bending?

Answer: C) Ligamentum flavum

Ligamentum flavum (elastic) connects adjacent laminae, limits abrupt flexion, and assists in straightening the column after flexing.

150. Per the gait lecture, inability to 'unlock' the extended knee implicates which muscle?

Answer: C) Popliteus

Popliteus unlocks the screw-home-locked knee to allow flexion; its dysfunction prevents unlocking.

151. The 1st carpometacarpal (thumb) joint, which allows opposition, is which synovial type?

Answer: C) Saddle

The thumb CMC joint (trapezium–base of 1st metacarpal) is a saddle joint, permitting the wide range that underlies opposition (prime mover: opponens pollicis).

Inside the examiner's head — what to drill: (1) deficit→structure reasoning (know the lost AND the spared function); (2) look-alike lesion pairs (AIN vs median, deep-ulnar vs whole-ulnar, tibial vs common fibular, spiral-groove vs axillary radial, ulnar paradox); (3) joint-type traps (wrist=condyloid not hinge, radio-ulnar=pivot, thumb CMC & SC=saddle); (4) compartment→nerve→artery→action grouping; (5) histology cell/stain ID (plasma-cell clock-face, reticular silver+, collagen types); (6) development eponyms (Poland, scaphocephaly, sclerotome, clubfoot); (7) spine ligaments & craniovertebral YES/NO movements. The professor recycles these concepts as fresh vignettes — reason, don't memorise the wording.